

[illegible]

TRAN NORDESTINA Logística S.A.																										FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS			
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																										NUMERAÇÃO DA TLSA		PÁGINA	
TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)																										FTN-MVP10-NS4D-0643-001-REV02		2 / 2	
MVP 10 - KM 450+000 A KM 500+789.330																										CONTRATO		REV.	
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																												00	

NOTA DE SERVIÇO - FERROVIA																																							
LADO ESQUERDO																		EIXO						LADO DIREITO															
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca	Cota Projeto	Cota Terreno	Cota Vermelha	Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset	
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)					Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)
- 11.317	89.815	- 5.049	93.993											- 3.550	94.993	453+520.00	95.099	90.028	5.072	9,548,560.163	526,977.929	3.550	94.993													5.049	93.993	11.284	89.837
- 11.320	89.895	- 5.049	94.075											- 3.550	95.075	453+540.00	95.181	90.105	5.076	9,548,570.572	526,960.853	3.550	95.075												5.049	94.075	11.296	89.910	
- 11.328	89.971	- 5.049	94.157											- 3.550	95.156	453+560.00	95.263	90.183	5.079	9,548,581.683	526,944.225	3.550	95.156												5.049	94.157	11.291	89.995	
- 11.270	90.091	- 5.049	94.239											- 3.550	95.238	453+580.00	95.345	90.263	5.082	9,548,593.567	526,928.141	3.550	95.238												5.049	94.239	11.297	90.074	
- 11.186	90.229	- 5.049	94.320											- 3.550	95.320	453+600.00	95.426	90.332	5.094	9,548,606.230	526,912.663	3.550	95.320												5.049	94.320	11.500	90.020	
- 11.125	90.351	- 5.049	94.402											- 3.550	95.402	453+620.00	95.508	90.430	5.078	9,548,619.642	526,897.830	3.550	95.402												5.049	94.402	11.501	90.101	
- 11.104	90.447	- 5.049	94.484											- 3.550	95.483	453+640.00	95.590	90.517	5.073	9,548,633.772	526,883.678	3.550	95.483												5.049	94.484	11.503	90.181	
- 11.111	90.524	- 5.049	94.566											- 3.550	95.565	453+660.00	95.672	90.592	5.080	9,548,648.583	526,870.241	3.550	95.565												5.049	94.566	11.517	90.254	
- 11.083	90.625	- 5.049	94.647											- 3.550	95.647	453+680.00	95.753	90.683	5.071	9,548,664.041	526,857.553	3.550	95.647												5.049	94.647	11.520	90.334	
- 11.241	90.601	- 5.049	94.729											- 3.550	95.729	453+700.00	95.835	90.602	5.233	9,548,680.106	526,845.645	3.550	95.729												5.049	94.729	11.853	90.193	
- 11.880	90.257	- 5.049	94.811											- 3.550	95.811	453+720.00	95.917	90.381	5.536	9,548,696.741	526,834.545	3.550	95.811												5.049	94.811	12.274	89.995	
- 12.651	89.825	- 5.049	94.893											- 3.550	95.892	453+740.00	95.999	90.040	5.958	9,548,713.904	526,824.281	3.550	95.892												5.049	94.893	12.688	89.800	
- 12.811	89.800	- 5.049	94.975											- 3.550	95.974	453+760.00	96.081	90.000	6.081	9,548,731.553	526,814.878	3.550	95.974												5.049	94.975	12.811	89.800	
- 12.588	90.030	- 5.049	95.056											- 3.550	96.056	453+780.00	96.162	90.000	6.162	9,548,749.642	526,806.350	3.550	96.056												5.049	95.056	12.934	89.800	
- 11.374	90.921	- 5.049	95.138											- 3.550	96.138	453+800.00	96.244	90.762	5.482	9,548,768.095	526,798.641	3.550	96.138												5.049	95.138	12.375	90.255	
- 10.015	91.909	- 5.049	95.220											- 3.550	96.219	453+820.00	96.326	91.462	4.864	9,548,786.835	526,791.658	3.550	96.219												5.049	95.220	11.860	90.679	
- 8.968	92.689	- 5.049	95.302											- 3.550	96.301	453+840.00	96.408	92.237	4.170	9,548,805.798	526,785.302	3.550	96.301												5.049	95.302	11.013	91.326	
- 8.353	93.181	- 5.049	95.383											- 3.550	96.383	453+860.00	96.489	92.751	3.739	9,548,824.927	526,779.467	3.550	96.383												5.049	95.383	10.223	91.934	
- 8.301	93.298	- 5.049	95.465											- 3.550	96.465	453+880.00	96.571	92.849	3.722	9,548,844.178	526,774.047	3.550	96.465												5.049	95.465	10.354	91.928	
- 8.758	93.075	- 5.049	95.547											- 3.550	96.546	453+900.00	96.653	92.592	4.061	9,548,863.513	526,768.931	3.550	96.546												5.049	95.547	11.032	91.558	
- 9.295	92.798	- 5.049	95.629											- 3.550	96.628	453+920.00	96.735	92.228	4.507	9,548,882.897	526,764.008	3.550	96.628												5.049	95.629	12.292	90.800	
- 10.085	92.353	- 5.049	95.710											- 3.550	96.710	453+940.00	96.816	92.000	4.816	9,548,902.302	526,759.165	3.550	96.710												5.049	95.710	11.025	91.727	
- 10.513	92.150	- 5.049	95.792											- 3.550	96.792	453+960.00	96.898	92.000	4.898	9,548,921.709	526,754.329	3.550	96.792												5.049	95.792	11.038	91.800	
- 10.890	91.980	- 5.049	95.874											- 3.550	96.874	453+980.00	96.980	92.162	4.818	9,548,941.115	526,749.494	3.550	96.874												5.049	95.874	10.050	92.540	
- 10.457	92.351	- 5.049	95.956											- 3.550	96.955	454+000.00	97.062	92.891	4.171	9,548,960.522	526,744.658	3.550	96.955												5.049	95.956	9.049	93.289	
- 9.416	93.126	- 5.049	96.038											- 3.550	97.037	454+020.00	97.144	93.462	3.682	9,548,979.928	526,739.822	3.550	97.037												5.049	96.038	8.775	93.554	
- 8.668	93.707	- 5.049	96.119											- 3.550	97.119	454+040.00	97.225	93.981	3.244	9,548,999.335	526,734.986	3.550	97.119												5.049	96.119	8.528	93.800	
- 8.628	93.815	- 5.049	96.201											- 3.550	97.201	454+060.00	97.307	93.875	3.432	9,549,018.741	526,730.150	3.550	97.201												5.049	96.201	9.173	93.452	
- 8.427	94.031	- 5.049	96.283											- 3.550	97.282	454+080.00	97.389	93.607	3.782	9,549,038.148	526,725.314	3.550	97.282												5.049	96.283	10.274	92.800	
- 8.375	94.148	- 5.049	96.365											- 3.550	97.364	454+100.00	97.471	93.665	3.806	9,549,057.555	526,720.478	3.550	97.364												5.049	96.365	10.396	92.800	
- 8.362	94.238	- 5.049	96.446											- 3.550	97.446	454+120.00	97.552	93.710	3.842	9,549,076.961	526,715.643	3.550	97.446												5.049	96.446	10.600	92.746	
- 9.112	93.820	- 5.049	96.528											- 3.550	97.528	454+140.00	97.634	93.428	4.206	9,549,096.368	526,710.807	3.550	97.528												5.049	96.528	11.445	92.265	
- 10.149	93.210	- 5.049	96.610											- 3.550	97.609	454+160.00	97.716	92.097	5.619	9,549,115.774	526,705.971	3.550	97.609												5.049	96.610	12.126	91.892	
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Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)					Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	
- 5.364	99.767	- 5.049	99.452											- 3.550	100.452	455+040.00	100.558	100.170	0.388	9,549,859.249	526,260.990	3.550	100.452											5.049	99.452	5.765	100.168			
- 30.221	100.685													- 29.250	99.714	455+060.00	100.592	102.248	- 1.656	9,549,876.323	526,250.574	29.250	99.714														30.845	101.309		
- 31.904	102.402													- 29.250	99.748	455+080.00	100.625	103.955	- 3.329	9,549,893.396	526,240.158	29.250	99.748															31.703	102.201	
- 33.076	103.608													- 29.250	99.782	455+100.00	100.659	104.243	- 3.584	9,549,910.470	526,229.743	29.250	99.782															31.237	101.768	
- 31.891	102.456													- 29.250	99.815	455+120.00	100.693	103.972	- 3.279	9,549,927.544	526,219.327	29.250	99.815																29.893	100.458
- 29.768	100.367													- 29.250	99.849	455+140.00	100.726	102.363	- 1.636	9,549,944.617	526,208.911	4.250	100.599															5.369	101.718	
- 5.343	99.948	- 5.049	99.654											- 3.550	100.654	455+160.00	100.760	99.872	0.888	9,549,961.691	526,198.495	3.550	100.654											5.049	99.654	5.885	99.097			
- 9.193	96.925	- 5.049	99.688											- 3.550	100.687	455+180.00	100.794	96.771	4.022	9,549,978.765	526,188.079	3.550	100.687											5.049	99.688	10.536	96.030			
- 14.545	93.391	- 5.049	99.721											- 3.550	100.721	455+200.00	100.827	93.705	7.122	9,549,995.838	526,177.663	3.550	100.721										5.049	99.721	14.775	93.237				
- 19.981	91.890	- 5.049	99.755									- 18.550	92.844	- 15.550	92.754	- 3.550	100.754	455+220.00	100.861	91.000	9.861	9,550,012.912	526,167.247	3.550	100.754	15.550	92.754	- 18.550	92.844					5.049	99.755	21.617	90.800			
- 19.905	91.975	- 5.049	99.789									- 18.550	92.878	- 15.550	92.788	- 3.550	100.788	455+240.00	100.895	92.332	8.563	9,550,029.985	526,156.831	3.550	100.788	15.550	92.788	- 18.550	92.878					5.049	99.789	20.167	91.800			
- 19.032	92.590	- 5.049	99.822									- 18.550	92.912	- 15.550	92.822	- 3.550	100.822	455+260.00	100.928	92.632	8.296	9,550,047.059	526,146.415	3.550	100.822	15.550	92.822	- 18.550	92.912					5.049	99.822	19.691	92.151			
- 18.747	92.814	- 5.049	99.856									- 18.550	92.945	- 15.550	92.855	- 3.550	100.855	455+280.00	100.962	92.957	8.005	9,550,064.121	526,135.981	3.550	100.855	15.550	92.855	- 18.550	92.945					5.049	99.856	18.935	92.688			
- 18.819	92.800	- 5.049	99.890									- 18.550	92.979	- 15.550	92.889	- 3.550	100.889	455+300.00	100.996	93.052	7.944	9,550,081.121	526,125.444	3.550	100.889	15.550	92.889	- 18.550	92.979					5.049	99.890	18.709	92.873			
- 15.378	93.037	- 5.049	99.923											- 3.550	100.923	455+320.00	101.029	93.280	7.750	9,550,097.994	526,114.707	3.550	100.923											5.049	99.923	15.314	93.080			
- 14.808	93.451	- 5.049	99.957											- 3.550	100.956	455+340.00	101.063	93.583	7.480	9,550,114.675	526,103.674	3.550	100.956											5.049	99.957	14.913	93.381			
- 14.473	93.708	- 5.049	99.990											- 3.550	100.990	455+360.00	101.096	93.863	7.233	9,550,131.094	526,092.255	3.550	100.990										5.049	99.990	14.509	93.684				
- 14.233	93.902	- 5.049	100.024											- 3.550	101.024	455+380.00	101.130	94.307	6.823	9,550,147.209	526,080.411	3.550	101.024											5.049	100.024	13.850	94.157			
- 13.892	94.162	- 5.049	100.058											- 3.550	101.057	455+400.00	101.164	94.801	6.363	9,550,163.009	526,068.150	3.550	101.057											5.049	100.058	13.384	94.501			
- 13.861	94.217	- 5.049	100.091											- 3.550	101.091	455+420.00	101.197	95.000	6.197	9,550,178.483	526,055.479	3.550	101.091											5.049	100.091	12.908	94.852			
- 14.015	94.148	- 5.049	100.125											- 3.550	101.124	455+440.00	101.231	95.058	6.173	9,550,193.619	526,042.408	3.550	101.124											5.049	100.125	12.405	95.221			
- 13.466	94.547	- 5.049	100.159											- 3.550	101.158	455+460.00	101.265	95.235	6.029	9,550,208.409	526,028.945	3.550	101.158											5.049	100.159	11.768	95.679			
- 12.770	95.045	- 5.049	100.192											- 3.550	101.192	455+480.00	101.298	95.655	5.644	9,550,222.840	526,015.099	3.550	101.192											5.049	100.192	11.311	96.018			
- 10.857	96.354	- 5.049	100.226											- 3.550	101.225	455+500.00	101.332	96.724	4.608	9,550,236.905	526,000.880	3.550	101.225											5.049	100.226	10.002	96.924			
- 9.096	97.561	- 5.049	100.260											- 3.550	101.259	455+520.00	101.366	98.139	3.227	9,550,250.594	525,986.300	3.550	101.259											5.049	100.260	7.818	98.414			
- 8.777	97.808	- 5.049	100.293											- 3.550	101.293	455+540.00	101.399	98.691	2.708	9,550,263.942	525,971.407	3.550	101.293											5.049	100.293	7.013	98.984			
- 9.360	97.453	- 5.049	100.327											- 3.550	101.326	455+560.00	101.433	98.418	3.015	9,550,277.028	525,956.282	3.550	101.326											5.049	100.327	7.267	98.848			
- 9.635	97.304	- 5.049	100.360											- 3.550	101.360	455+580.00	101.466	98.000	3.466	9,550,289.937	525,941.006	3.550	101.360										5.049	100.360	8.187	98.268				
- 7.955	98.457	- 5.049	100.394											- 3.550	101.394	455+600.00	101.500	98.000	3.500	9,550,302.757	525,925.656	3.550	101.394										5.049	100.394	8.756	97.923				
- 5.345	100.231	- 5.049	100.428											- 3.550	101.427	455+620.00	101.534	99.562	1.972	9,550,315.562	525,910.292	3.550	101.427											5.049	100.428	8.332	98.239			
- 6.502	101.914	- 5.049	100.461											- 3.550	101.461	455+640.00	101.567	101.329	0.238	9,550,328.368	525,894.929	3.550																		

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- 7.273	114.032															- 4.250	111.009	458+120.00	111.136	114.497	- 3.360	9,552,333.366	524,695.584	4.250	111.009													8.080	114.839						
- 5.092	110.159	- 5.049	110.187													- 3.550	111.187	458+140.00	111.293	110.753	0.540	9,552,353.108	524,692.381	3.550	111.187							5.049	110.187	6.025	111.163										
- 7.271	114.344															- 4.250	111.323	458+160.00	111.450	114.960	- 3.510	9,552,372.850	524,689.178	4.250	111.323													7.972	115.045						
- 13.662	114.711											- 11.266	112.316	- 5.266	112.496	- 4.250	111.480	458+180.00	111.607	115.000	- 3.393	9,552,392.592	524,685.975	4.250	111.480	5.513	112.742	11.513	112.562										14.837	115.886					
- 4.530	111.917															- 4.250	111.637	458+200.00	111.764	112.368	- 0.604	9,552,412.334	524,682.772	4.250	111.637														5.076	112.463					
- 7.013	109.506	- 5.049	110.815													- 3.550	111.815	458+220.00	111.921	110.258	1.663	9,552,432.075	524,679.569	3.550	111.815									5.049	110.815	5.906	110.244								
- 4.617	112.317															- 4.250	111.950	458+240.00	112.078	112.933	- 0.855	9,552,451.817	524,676.366	4.250	111.950															5.453	113.153				
- 5.092	112.950															- 4.250	112.107	458+260.00	112.235	113.509	- 1.274	9,552,471.559	524,673.163	4.250	112.107															5.899	113.756				
- 5.233	111.469	- 5.049	111.286													- 3.550	112.285	458+280.00	112.392	112.086	0.306	9,552,491.301	524,669.960	3.550	112.285									5.049	111.286	6.181	112.417								
- 6.902	110.208	- 5.049	111.443													- 3.550	112.442	458+300.00	112.549	110.935	1.614	9,552,511.043	524,666.757	3.550	112.442											5.049	111.443	5.561	111.101						
- 8.421	109.352	- 5.049	111.600													- 3.550	112.599	458+320.00	112.706	109.952	2.754	9,552,530.785	524,663.554	3.550	112.599											5.049	111.600	7.457	109.994						
- 10.713	107.981	- 5.049	111.757													- 3.550	112.756	458+340.00	112.863	108.100	4.762	9,552,550.526	524,660.351	3.550	112.756											5.049	111.757	10.978	107.804						
- 8.949	109.314	- 5.049	111.914													- 3.550	112.913	458+360.00	113.020	109.772	3.247	9,552,570.268	524,657.148	3.550	112.913												5.049	111.914	8.174	109.830					
- 7.140	110.677	- 5.049	112.070													- 3.550	113.070	458+380.00	113.176	111.372	1.804	9,552,590.010	524,653.945	3.550	113.070												5.049	112.070	5.852	111.535					
- 5.204	112.382	- 5.049	112.227													- 3.550	113.227	458+400.00	113.333	113.247	0.086	9,552,609.752	524,650.742	3.550	113.227												5.049	112.227	6.651	113.829					
- 5.100	112.435	- 5.049	112.384													- 3.550	113.384	458+420.00	113.490	113.302	0.188	9,552,629.494	524,647.539	3.550	113.384											5.049	112.384	6.581	113.916						
- 5.136	112.483	- 5.049	112.541													- 3.550	113.541	458+440.00	113.647	113.216	0.431	9,552,649.236	524,644.336	3.550	113.541											5.049	112.541	6.148	113.640						
- 4.679	113.391															- 4.250	113.677	458+460.00	113.804	114.285	- 0.481	9,552,668.977	524,641.132	4.250	113.677																	5.499	114.925		
- 4.539	114.123															- 4.250	113.834	458+480.00	113.961	115.469	- 1.508	9,552,688.719	524,637.929	4.250	113.834																	7.140	116.724		
- 5.241	113.330															- 4.250	113.990	458+500.00	114.118	114.525	- 0.407	9,552,708.461	524,634.726	4.250	113.990																	5.478	115.218		
- 5.871	112.621	- 5.049	113.169													- 3.550	114.168	458+520.00	114.275	113.296	0.979	9,552,728.203	524,631.523	3.550	114.168										5.049	113.169	5.406	113.525							
- 5.838	112.800	- 5.049	113.326													- 3.550	114.325	458+540.00	114.432	113.276	1.155	9,552,747.945	524,628.320	3.550	114.325											5.049	113.326	5.126	113.403						
- 7.641	111.755	- 5.049	113.483													- 3.550	114.482	458+560.00	114.589	112.741	1.848	9,552,767.687	524,625.117	3.550	114.482											5.049	113.483	5.647	113.084						
- 9.309	110.800	- 5.049	113.640													- 3.550	114.639	458+580.00	114.746	111.887	2.859	9,552,787.429	524,621.914	3.550	114.639												5.049	113.640	6.907	112.401					
- 10.629	110.077	- 5.049	113.797													- 3.550	114.796	458+600.00	114.903	111.039	3.864	9,552,807.170	524,618.711	3.550	114.796											5.049	113.797	8.541	111.469						
- 11.912	109.378	- 5.049	113.954													- 3.550	114.953	458+620.00	115.060	110.295	4.765	9,552,826.912	524,615.508	3.550	114.953											5.049	113.954	10.286	110.463						
- 13.073	108.762	- 5.049	114.110													- 3.550	115.110	458+640.0																											

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																														FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS										NUMERAÇÃO DA TISA		PÁGINA	
TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)																																								FTN-MVP10-NS4D-0643-001-REV02		2 / 2	
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NOTA DE SERVIÇO - FERROVIA																																											
LADO ESQUERDO																		EIXO						LADO DIREITO																			
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca	Cota Projeto	Cota Terreno	Cota Vermelha	Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset			
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Norte (m)	Leste (m)					Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)		
- 17.673	127.189											- 15.131	124.647	- 9.131	124.827	- 4.250	119.946	459+880.00	120.073	129.988	- 9.915	9,553,986.496	524,234.096	14.250	119.646	22.250	127.646	28.250	127.466	28.659	127.875	34.659	127.695							39.150	132.186		
- 7.886	123.622															- 4.250	119.986	459+900.00	120.113	125.516	- 5.403	9,553,997.024	524,217.092	14.250	119.686	22.250	127.686	28.250	127.506	30.726	129.982	36.726	129.802							39.236	132.312		
- 8.705	124.481															- 4.250	120.025	459+920.00	120.153	126.605	- 6.452	9,554,006.952	524,199.732	14.250	119.725	22.250	127.725	28.250	127.545	31.170	130.465	37.170	130.285							39.378	132.493		
- 5.682	121.497															- 4.250	120.065	459+940.00	120.193	123.208	- 3.016	9,554,016.268	524,182.035	14.250	119.765	22.250	127.765	28.250	127.585	30.780	130.115	36.780	129.935							38.920	132.075		
- 6.837	118.380															- 4.250	120.105	459+960.00	120.232	120.731	- 0.499	9,554,024.961	524,164.024	14.250	119.805														19.286	124.841			
- 7.167	117.754	- 5.049	119.166													- 3.550	120.166	459+980.00	120.272	119.254	- 1.018	9,554,033.021	524,145.721	3.550	120.166											5.049	119.166	5.880	119.997				
- 5.993	120.150	- 5.049	119.206													- 3.550	120.205	460+000.00	120.312	120.058	- 0.254	9,554,040.440	524,127.149	3.550	120.205													5.049	119.206	5.484	119.641		
- 6.146	122.120															- 4.250	120.224	460+020.00	120.352	123.000	- 2.648	9,554,047.275	524,108.354	4.250	120.224														6.992	122.966			
- 5.515	119.751	- 5.049	119.285													- 3.550	120.285	460+040.00	120.391	120.212	- 0.179	9,554,053.627	524,089.390	3.550	120.285											5.049	119.285	5.808	120.044				
- 10.147	119.273	- 10.049	119.175													- 8.550	120.174	460+060.00	120.431	120.310	- 0.121	9,554,059.604	524,070.304	3.550	120.324													5.274	119.175	6.899	120.800		
- 9.929	120.873															- 9.250	120.193	460+080.00	120.471	120.970	- 0.499	9,554,065.315	524,051.137	4.250	120.343														4.581	120.674			
- 11.254	118.451	- 10.049	119.254													- 8.550	120.254	460+100.00	120.510	118.891	- 1.619	9,554,070.870	524,031.924	3.550	120.404											5.274	119.254	6.738	118.279				
- 16.322	115.112	- 10.049	119.294													- 8.550	120.294	460+120.00	120.550	115.899	- 4.651	9,554,076.377	524,012.697	3.550	120.444													5.274	119.294	10.026	116.127		
- 16.445	115.070	- 10.049	119.334													- 8.550	120.333	460+140.00	120.590	115.541	- 5.049	9,554,081.883	523,993.470	3.550	120.483													5.274	119.334	11.108	115.445		
- 12.369	117.827	- 10.049	119.374													- 8.550	120.373	460+160.00	120.630	117.821	- 2.808	9,554,087.389	523,974.243	3.550	120.523													5.274	119.374	9.614	116.480		
- 10.720	121.862															- 9.250	120.392	460+180.00	120.669	121.660	- 0.991	9,554,092.896	523,955.016	4.250	120.542														4.412	120.704			
- 10.826	122.007															- 9.250	120.432	460+200.00	120.709	123.829	- 3.120	9,554,098.402	523,935.789	4.250	120.582														7.263	123.595			
- 11.311	122.532															- 9.250	120.471	460+220.00	120.749	124.364	- 3.615	9,554,103.908	523,916.562	4.250	120.621														8.532	124.903			
- 10.539	121.800															- 9.250	120.511	460+240.00	120.788	125.127	- 4.338	9,554,109.414	523,897.335	4.250	120.661	10.250	126.661	16.250	126.481										20.753	130.984			
- 24.199	129.320											- 17.674	122.794	- 11.674	122.974	- 9.250	120.551	460+260.00	120.828	125.703	- 4.875	9,554,114.920	523,878.107	4.250	120.701														11.312	127.762			
- 25.640	130.800											- 18.456	123.617	- 12.456	123.797	- 9.250	120.590	460+280.00	120.868	129.353	- 8.485	9,554,120.426	523,858.880	4.250	120.740	6.283	122.774	12.283	122.594	18.283	128.594	24.283	128.414	30.283	134.414	36.283	134.234			41.252	139.202		
- 23.848	129.048											- 18.499	123.700	- 12.499	123.880	- 9.250	120.630	460+300.00	120.908	128.008	- 7.100	9,554,125.932	523,839.653	4.250	120.780	9.163	125.693	15.163	125.513	21.163	131.513	27.163	131.333							30.086	134.256		
- 19.888	125.128											- 16.744	121.984	- 10.744	122.164	- 9.250	120.670	460+320.00	120.947	127.341																							

TRANSPORTAÇÃO LOGÍSTICA S.A.																												FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS															
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																												NUMERAÇÃO DA TISA										PÁGINA					
TRECHO MISSÃO VELHA (CE) - PORTO DO PECEM (CE)																												FTN-MVP10-NS4D-0643-001-REV02										2 / 2					
MVP 10 - KM 450+000 A KM 500+789.330																												CONTRATO										REV.					
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NOTA DE SERVIÇO - FERROVIA																																											
LADO ESQUERDO																		LADO DIREITO																									
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca		Cota		Cota		Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset	
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Projeto	Terreno	Vermeilha	Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)			
- 22.983	132.156											- 19.279	128.451	- 13.279	128.631	- 9.250	124.603	462+300.00	124.880	132.000	- 7.120	9,555,590.657	522,587.711	4.250	124.753	8.527	129.030	14.527	128.850													18.252	132.575
- 23.194	132.406											- 18.707	127.920	- 12.707	128.100	- 9.250	124.642	462+320.00	124.920	131.997	- 7.077	9,555,605.189	522,573.970	4.250	124.792	6.695	127.237	12.695	127.057													16.363	130.725
- 21.728	130.980											- 18.137	127.389	- 12.137	127.569	- 9.250	124.682	462+340.00	124.960	129.996	- 5.037	9,555,619.720	522,560.228	4.250	124.832	5.791	126.373	11.791	126.193													14.242	128.644
- 19.541	128.833											- 16.952	126.244	- 10.952	126.424	- 9.250	124.722	462+360.00	124.999	128.186	- 3.187	9,555,634.252	522,546.487	4.250	124.872	4.867	125.489	10.867	125.309													12.418	126.860
- 10.009	125.520															- 9.250	124.761	462+380.00	125.039	125.710	- 0.671	9,555,648.784	522,532.745	4.250	124.911																4.676	125.338	
- 12.708	122.050	- 10.049	123.823													- 8.550	124.822	462+400.00	125.079	123.618	1.460	9,555,663.316	522,519.004	3.550	124.972									5.274	123.823	5.828	123.454						
- 17.047	119.197	- 10.049	123.862													- 8.550	124.862	462+420.00	125.118	121.392	3.727	9,555,677.847	522,505.262	3.550	125.012									5.274	123.862	8.574	121.662						
- 17.703	118.800	- 10.049	123.902													- 8.550	124.902	462+440.00	125.158	119.983	5.175	9,555,692.395	522,491.538	3.550	125.052									5.274	123.902	10.753	120.250						
- 17.187	119.183	- 10.049	123.942													- 8.550	124.941	462+460.00	125.198	119.000	6.198	9,555,707.026	522,477.902	3.550	125.091									5.274	123.942	12.987	118.800						
- 9.382	121.244	- 5.049	124.132													- 3.550	125.132	462+480.00	125.238	121.110	4.128	9,555,721.819	522,464.443	3.550	125.132									5.049	124.132	11.117	120.087						
- 6.177	123.427	- 5.049	124.179													- 3.550	125.178	462+500.00	125.285	123.323	1.962	9,555,736.849	522,451.249	3.550	125.178									5.049	124.179	7.015	122.868						
- 5.507	124.690	- 5.049	124.232													- 3.550	125.231	462+520.00	125.338	124.506	0.832	9,555,752.187	522,438.414	3.550	125.231									5.049	124.232	5.550	123.898						
- 5.063	124.306	- 5.049	124.292													- 3.550	125.291	462+540.00	125.398	124.129	1.269	9,555,767.894	522,426.035	3.550	125.291									5.049	124.292	6.361	123.417						
- 7.170	122.944	- 5.049	124.358													- 3.550	125.358	462+560.00	125.464	122.661	2.803	9,555,784.028	522,414.217	3.550	125.358									5.049	124.358	8.886	121.800						
- 5.466	124.154	- 5.049	124.431													- 3.550	125.431	462+580.00	125.537	124.152	1.385	9,555,800.620	522,403.052	3.550	125.431									5.049	124.431	6.088	123.738						
- 5.057	126.296															- 4.250	125.489	462+600.00	125.617	126.213	- 0.596	9,555,817.653	522,392.573	4.250	125.489																		
- 6.726	128.052															- 4.250	125.576	462+620.00	125.703	127.604	- 1.901	9,555,835.098	522,382.795	4.250	125.576																		
- 7.301	128.720															- 4.250	125.669	462+640.00	125.796	127.957	- 2.160	9,555,852.927	522,373.736	4.250	125.669																		
- 6.467	127.986															- 4.250	125.769	462+660.00	125.896	127.154	- 1.258	9,555,871.110	522,365.410	4.250	125.769																		
- 5.937	127.562															- 4.250	125.875	462+680.00	126.003	126.827	- 0.825	9,555,889.617	522,357.831	4.250	125.875																		
- 6.062	127.800															- 4.250	125.988	462+700.00	126.116	127.000	- 0.884	9,555,908.417	522,351.011	4.250	125.988																		
- 7.122	128.980															- 4.250	126.108	462+720.00	126.235	129.142	- 2.907	9,555,927.479	522,344.963	4.250	126.108																		
- 13.923	129.727											- 10.506	126.310	- 4.506	126.490	- 4.250	126.234	462+740.00	126.362	130.280	- 3.918	9,555,946.770	522,339.688	4.250	126.234	5.130	127.114	11.130	126.934														
- 5.761	127.879															- 4.250	126.367	462+760.00	126.495	129.513	- 3.018	9,555,966.239	522,335.116	4.250	126.367																		
- 5.026	125.990															- 4.250	126.507	462+780.00	126.635	127.293	- 0.659	9,555,985.839	522,331.141	4.250	126.507																		
- 5.650	128.053															- 4.250	126.654	462+800.00	126.781	128.261	- 1.480	9,556,005.532	522,327.651	4.250	126.654																		
- 13.458	129.834											- 12.907	129.284	- 6.907	129.464	- 4.250	126.807	462+820.00																									

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																																FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS				NUMERAÇÃO DA TISA		PÁGINA			
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																																FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS				FTM-MVP10-NS4D-0643-001-REV02		2 / 2			
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																																FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS				CONTRATO		REV.			
00																																FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS									
NOTA DE SERVIÇO - FERROVIA																																									
LADO ESQUERDO																EIXO						LADO DIREITO																			
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca		Cota		Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset	
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)		Projeto	Terreno	Vermelha	Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)		
-18.773	147.922											-15.652	144.800	-11.652	144.920	-4.250	137.519	464+060.00	137.646	145.892	-8.246	9,557,142.500	521,855.506	4.250	137.519	9.139	142.408	-13.139	142.288											14.759	143.908
-17.853	147.175											-14.809	144.131	-10.809	144.251	-4.250	137.692	464+080.00	137.819	145.647	-7.828	9,557,157.865	521,842.702	4.250	137.692	8.790	142.232	-12.790	142.112											14.493	143.815
-16.545	146.040											-13.406	142.901	-9.406	143.021	-4.250	137.865	464+100.00	137.993	145.011	-7.018	9,557,173.230	521,829.899	4.250	137.865	7.924	141.539	-11.924	141.419											14.049	143.544
-15.135	144.803											-11.486	141.155	-7.486	141.275	-4.250	138.038	464+120.00	138.166	143.982	-5.817	9,557,188.594	521,817.095	4.250	138.038	6.347	140.135	-10.347	140.015											13.024	142.692
-12.959	142.800											-9.491	139.332	-5.491	139.452	-4.250	138.211	464+140.00	138.339	142.000	-3.661	9,557,203.959	521,804.292	4.250	138.211	5.546	139.507	-9.546	139.387											12.120	141.961
-14.658	144.672											-11.223	141.237	-7.223	141.357	-4.250	138.384	464+160.00	138.512	144.739	-6.227	9,557,219.324	521,791.489	4.250	138.384	7.041	141.175	-11.041	141.055											14.340	144.354
-16.438	146.625											-12.964	143.152	-8.964	143.272	-4.250	138.557	464+180.00	138.685	146.499	-7.814	9,557,234.689	521,778.685	4.250	138.557	8.490	142.798	-12.490	142.678											15.605	145.793
-18.111	148.471											-14.315	144.675	-10.315	144.795	-4.250	138.731	464+200.00	138.858	147.499	-8.641	9,557,250.053	521,765.882	4.250	138.731	9.026	143.507	-13.026	143.387											15.803	146.163
-18.286	148.820											-14.340	144.873	-10.340	144.993	-4.250	138.904	464+220.00	139.031	147.653	-8.622	9,557,265.418	521,753.079	4.250	138.904	8.833	143.486	-12.833	143.366											15.618	146.152
-17.085	147.792											-13.500	144.207	-9.500	144.327	-4.250	139.077	464+240.00	139.204	146.990	-7.785	9,557,280.783	521,740.275	4.250	139.077	8.434	143.261	-12.434	143.141											15.138	145.844
-15.294	146.174											-11.951	142.830	-7.951	142.950	-4.250	139.250	464+260.00	139.377	145.321	-5.944	9,557,296.147	521,727.472	4.250	139.250	6.729	141.729	-10.729	141.609											12.719	143.599
-13.346	144.399											-10.048	141.101	-6.048	141.221	-4.250	139.423	464+280.00	139.551	144.098	-4.547	9,557,311.512	521,714.669	4.250	139.423	6.164	141.337	-10.164	141.217											12.747	143.800
-13.656	144.882											-11.082	142.309	-7.082	142.429	-4.250	139.596	464+300.00	139.724	145.582	-5.858	9,557,326.877	521,701.865	4.250	139.596	7.342	142.688	-11.342	142.568											14.300	145.526
-15.723	147.122											-13.043	144.442	-9.043	144.562	-4.250	139.769	464+320.00	139.897	147.352	-7.455	9,557,342.242	521,689.062	4.250	139.769	8.912	144.432	-12.912	144.312											15.486	146.885
-17.403	148.976											-14.442	146.014	-10.442	146.134	-4.250	139.942	464+340.00	140.070	149.290	-9.220	9,557,357.606	521,676.258	4.250	139.942	9.962	145.655	-13.962	145.535											16.605	148.177
-19.300	151.046											-16.055	147.801	-12.055	147.921	-4.250	140.115	464+360.00	140.243	150.436	-10.193	9,557,372.948	521,663.428	4.250	140.115	10.503	146.368	-14.503	146.248											17.135	148.881
-24.651	152.449							-20.948	148.746	-16.948	148.866	-16.250	148.169	-12.250	148.289	-4.250	140.289	464+380.00	140.416	150.773	-10.357	9,557,388.198	521,650.489	4.250	140.289	10.579	146.618	-14.579	146.498											17.286	149.204
-19.791	151.882											-15.989	148.080	-11.989	148.200	-4.250	140.462	464+400.00	140.589	150.570	-9.981	9,557,403.280	521,637.353	4.250	140.462	10.203	146.415	-14.203	146.295											16.708	148.800
-19.488	151.753											-15.090	147.355	-11.090	147.475	-4.250	140.635	464+420.00	140.762	151.657	-10.894	9,557,418.113	521,623.938	4.250	140.635	11.077	147.462	-15.077	147.342											18.468	150.733
-25.155	153.472							-21.953	150.271	-17.953	150.391	-16.250	148.688	-12.250	148.808	-4.250	140.808	464+440.00	140.935	152.726	-11.791	9,557,432.611	521,610.162	4.250	140.808	12.250	148.808	-16.250	148.688	-16.850	149.288	-20.850	149.168							23.495	151.813
-26.298	154.789							-23.417	151.908	-19.417	152.028	-16.250	148.861	-12.250	148.981	-4.250	140.981	464+460.00	141.109	154.040	-12.931	9,557,446.689	521,595.957	4.250	140.981	12.250	148.981	-16.250	148.861	-18.287	150.898	-22.287	150.778							24.657	153.148
-27.428	156.092							-24.943	153.608	-20.943	153.728	-16.250	149.034	-12.250	149.154	-4.250	141.154	464+480.00	141.282	155.585	-14.303	9,557,460.306	521,581.310	4.250	141.154	12.250	149.154	-16.250	149.034	-20.147	152.931	-24.147	152.811							26.282	154.946
-28.832	157.669							-26.549	155.386	-22.549	155.506	-16.250	149.207	-12.250	149.327	-4.250	141.327	464+500.00	141.455	157.604	-16.149	9,557,473.447	521,566.234	4.250	141.327	12.250	149.327	-16.250	149.207	-21.805	154.763	-25.805	154.643							27.771	156.608
-34.671	159.562		</																																						

NOTA DE SERVIÇO - FERROVIA																																																
LADO ESQUERDO														EIXO						LADO DIREITO																												
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca		Cota		Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset								
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Estaca	Cota Projeto	Cota Terreno	Cota Vermelha	Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)							
- 22.816	135.256											- 20.870	133.311	- 14.870	133.491	- 14.250	132.870	467+520.00	133.298	136.808	- 3.510	9,559,761.051	520,926.595	10.250	132.990	11.554	134.295	17.554	134.115													19.960	136.520					
- 15.943	134.425															- 14.250	132.731	467+540.00	133.159	135.554	- 2.395	9,559,773.619	520,911.038	10.250	132.851																			13.298	135.899			
- 14.458	132.800															- 14.250	132.592	467+560.00	133.020	134.000	- 0.980	9,559,785.868	520,895.228	10.250	132.712																				11.662	134.124		
- 15.741	133.944															- 14.250	132.453	467+580.00	132.881	135.476	- 2.595	9,559,797.699	520,879.103	10.250	132.573																				13.441	135.764		
- 22.916	134.800											- 20.817	132.701	- 14.817	132.881	- 14.250	132.314	467+600.00	132.742	136.664	- 3.922	9,559,809.008	520,862.609	10.250	132.434	11.760	133.944	17.760	133.764															21.277	137.281			
- 24.468	136.213											- 21.640	133.385	- 15.640	133.565	- 14.250	132.175	467+620.00	132.603	136.947	- 4.344	9,559,819.703	520,845.710	10.250	132.295	12.412	134.457	18.412	134.277															21.579	137.445			
- 25.134	136.740											- 22.636	134.243	- 16.636	134.423	- 14.250	132.036	467+640.00	132.464	137.000	- 4.536	9,559,829.751	520,828.419	10.250	132.156	12.664	134.570	18.664	134.390															21.272	136.998			
- 24.708	136.175											- 22.280	133.747	- 16.280	133.927	- 14.250	131.897	467+660.00	132.325	136.605	- 4.280	9,559,839.138	520,810.760	10.250	132.017	12.240	134.007	18.240	133.827															20.942	136.529			
- 24.191	135.519											- 21.463	132.791	- 15.463	132.971	- 14.250	131.758	467+680.00	132.185	136.038	- 3.852	9,559,847.863	520,792.765	10.250	131.878	11.399	133.027	17.399	132.847															20.293	135.741			
- 23.692	134.881											- 20.670	131.859	- 14.670	132.039	- 14.250	131.619	467+700.00	132.046	135.425	- 3.379	9,559,855.996	520,774.494	10.250	131.739	10.614	132.103	16.614	131.923															19.746	135.055			
- 22.963	134.013											- 20.382	131.432	- 14.382	131.612	- 14.250	131.479	467+720.00	131.907	134.771	- 2.864	9,559,863.636	520,756.011	10.250	131.599	10.830	132.180	16.830	132.000															19.343	134.513			
- 15.690	132.773															- 14.250	131.333	467+740.00	131.760	133.681	- 1.921	9,559,870.885	520,737.372	10.250	131.453																				12.460	133.663		
- 4.406	131.633															- 4.250	131.477	467+760.00	131.605	132.078	- 0.473	9,559,877.850	520,718.624	10.250	131.297																				11.239	132.286		
- 6.687	129.242	- 5.049	130.334													- 3.550	131.334	467+780.00	131.440	129.189	- 2.251	9,559,884.639	520,699.811	3.550	131.334																			5.049	130.334	5.851	129.800	
- 17.045	133.634															- 14.250	130.839	467+800.00	131.267	133.040	- 1.773	9,559,891.359	520,680.974	10.250	130.959																						10.380	131.089
- 24.917	135.143											- 23.199	133.426	- 17.199	133.606	- 14.250	130.656	467+820.00	131.084	133.046	- 1.962	9,559,898.075	520,662.135	10.250	130.776	11.672	132.198	17.672	132.018																	18.504	132.850	
- 26.197	136.231											- 22.701	132.736	- 16.701	132.916	- 14.250	130.465	467+840.00	130.892	135.490	- 4.598	9,559,904.791	520,643.297	10.250	130.585	11.154	131.488	17.154	131.308																19.630	133.784		
- 27.620	137.454											- 24.555	134.389	- 18.555	134.569	- 14.250	130.264	467+860.00	130.691	136.620	- 5.929	9,559,911.507	520,624.458	10.250	130.384	12.673	132.807	18.673	132.627																20.733	134.687		
- 28.945	138.569											- 26.152	135.776	- 20.152	135.956	- 14.250	130.054	467+880.00	130.481	137.531	- 7.050	9,559,918.187	520,605.606	10.250	130.174	14.316	134.240	20.316	134.060																22.477	136.221		
- 30.264	139.669											- 27.509	136.914	- 21.509	137.094	- 14.250	129.835	467+900.00	130.263	138.544	- 8.281	9,559,924.729	520,586.707	10.250	129.955	15.693	135.398	21.693	135.218																23.834	137.359		
- 31.317	140.494											- 28.135	137.312	- 22.135	137.492	- 14.250	129.607	467+920.00	130.035	139.371	- 9.337	9,559,931.023	520,567.723	10.250	129.727	16.136	135.613	22.136	135.433																24.678	137.975		
- 38.040	140.800											- 28.250	137.190	- 22.250	137.370	- 14.250	129.370	467+940.00	129.798	139.859	- 10.061	9,559,936.956	520,548.624	10.250	129.490	15.924	135.164	21.924	134.984																24.893	137.954		
- 30.755	139.457											- 25.929	134.631	- 19.929	134.811	- 14.250	129.132	467+960.00	129.559	139.649	- 10.090	9,559,942.414	520,529.384	10.250	129.252	13.717	132.719	19.717	132.539															24.157	136.979			
- 27.970	136.433											- 23.023	131.487	- 17.023	131.667	- 14.250	128.893	467+980.00	129.321	137.317	- 7.997	9,559,947.282	520,509.986	10.250	129.013	10.997	129.760	16.997	129.580																21.723	134.306		
- 16.545	130.949															- 14.250	128.655	468+000.00	129.082	131.728	- 2.646	9,559,951.478	520,490.432	10.250	128.775																						12.211	130.735
- 4.591	129.057															- 4.250	128.716	468+020.00	128.843	129.568	- 0.724	9,559,954.989	520,470.744	10.250	128.536																						11.268	129.554
- 17.560	131.488															- 14.250	128.177	468+040.00	128.605	133.783	- 5.178	9,559,957.811	520,450.945	10.250	128.297	11.729	129.776	17.729	129.596																		22.176	134.043
- 15.385	129.073															- 14.250	127.939	468+060.00	128.366	131.945	- 3.579	9,559,959.941	520,431.060	10.250	128.059	12.579	130.388	18.579	130.208																	22.438	134.067	
- 4.775	128.525															- 4.250	128.000	468+080.00	128.127	129.312	- 1.184	9,559,961.432	520,411.116	10.250	127.820																					12.163	129.733	
- 7.914	124.873	- 5.049	126.783													- 3.550	127.782	468+100.00	127.889	125.452	- 2.437	9,559,962.395	520,391.140	3.550	127.782																				5.049	126.783	7.811	124.941
- 7.972	124.596	- 5.049	126.544													- 3.550	127.544	468+120.00	127.650	125.417	- 2.233	9,559,962.945	520,371.148	3.550	127.544																			5.049	126.544	6.620	125.497	
- 4.743	127.777															- 4.250	127.284	468+140.00	127.411	128.625	- 1.214	9,559,963.198	520,351.149	4.250	127.284																						6.051	129.084
- 13.460	130.075											- 12.396	129.011	- 6.396	129.191	- 4.250	127.045	468+160.00	127.173	131.845	- 4.672	9,559,963.271	520,331.149	4.250	127.045	7.824	130.619	13.824	130.439																15.953	132.569		
- 15.001	131.377											- 13.156	129.533</																																			

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																														FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS													
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																														NUMERAÇÃO DA TLSA				PÁGINA									
TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)																														FTN-MVP10-NS4D-0643-001-REV02				2 / 2									
MVP 10 - KM 450+000 A KM 500+789.330																														CONTRATO				REV.									
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																																		00									
NOTA DE SERVIÇO - FERROVIA																																											
LADO ESQUERDO																		LADO DIREITO																									
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca		Cota		Cota		Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset	
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)		Projeto	Terreno	Vermelha	Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)				
- 26.426	84.551	- 5.049	96.712									- 18.550	89.802	- 15.550	89.712	- 3.550	97.712	470+620.00	97.818	85.505	12.314	9,561,825.677	519,564.010	3.550	97.712	15.550	89.712	- 18.550	89.802									5.049	96.712	23.522	86.487		
- 28.601	82.863	- 5.049	96.474									- 18.550	89.563	- 15.550	89.473	- 3.550	97.473	470+640.00	97.580	83.495	14.084	9,561,839.298	519,578.653	3.550	97.473	15.550	89.473	- 18.550	89.563							5.049	96.474	25.162	85.156				
- 21.664	87.249	- 5.049	96.235									- 18.550	89.325	- 15.550	89.235	- 3.550	97.235	470+660.00	97.341	84.557	12.785	9,561,852.456	519,593.714	3.550	97.235	15.550	89.235	- 18.550	89.325							5.049	96.235	26.146	84.261				
- 9.894	92.767	- 5.049	95.996													- 3.550	96.996	470+680.00	97.102	90.713	6.390	9,561,865.210	519,609.119	3.550	96.996	15.550	88.996	- 18.550	89.086							5.049	95.996	22.989	86.127				
- 8.142	98.851	- 5.049	95.758													- 3.550	96.757	470+700.00	96.864	96.722	0.142	9,561,877.643	519,624.785	3.550	96.757									5.049	95.758	7.133	94.369						
- 5.939	98.187															- 4.250	96.498	470+720.00	96.625	99.018	- 2.393	9,561,889.841	519,640.635	4.250	96.498												6.629	98.877					
- 4.307	96.316															- 4.250	96.259	470+740.00	96.387	97.488	- 1.102	9,561,901.894	519,656.594	4.250	96.259												6.594	98.603					
- 4.889	95.594															- 4.250	96.020	470+760.00	96.148	96.579	- 0.432	9,561,913.895	519,672.594	4.250	96.020												5.438	97.208					
- 7.236	93.345	- 5.049	94.803													- 3.550	95.803	470+780.00	95.909	93.753	2.157	9,561,925.892	519,688.596	3.550	95.803									5.049	94.803	5.831	94.282						
- 12.824	89.385	- 5.049	94.568													- 3.550	95.568	470+800.00	95.674	92.190	3.484	9,561,937.890	519,704.598	3.550	95.568									5.049	94.568	7.326	93.050						
- 9.799	91.174	- 5.049	94.340													- 3.550	95.340	470+820.00	95.446	93.232	2.214	9,561,949.888	519,720.599	3.550	95.340									5.049	94.340	5.624	93.957						
- 9.933	90.864	- 5.049	94.120													- 3.550	95.119	470+840.00	95.226	92.842	2.384	9,561,961.885	519,736.601	3.550	95.119									5.049	94.120	5.742	93.658						
- 4.708	95.343															- 4.250	94.885	470+860.00	95.013	96.649	- 1.637	9,561,973.883	519,752.603	4.250	94.885												7.571	98.207					
- 4.324	94.753															- 4.250	94.679	470+880.00	94.807	95.973	- 1.166	9,561,985.881	519,768.605	4.250	94.679												7.174	97.603					
- 9.889	90.275	- 5.049	93.502													- 3.550	94.502	470+900.00	94.608	93.355	1.253	9,561,997.878	519,784.606	3.550	94.502									5.049	93.502	6.896	95.349						
- 20.200	85.300	- 5.049	93.311									- 18.550	86.400	- 15.550	86.310	- 3.550	94.310	470+920.00	94.417	89.666	4.750	9,562,009.876	519,800.608	3.550	94.310									5.049	93.311	7.697	91.545						
- 22.174	83.800	- 5.049	93.126									- 18.550	86.216	- 15.550	86.126	- 3.550	94.126	470+940.00	94.232	85.975	8.258	9,562,021.873	519,816.610	3.550	94.126									5.049	93.126	15.241	86.332						
- 21.130	84.319	- 5.049	92.950									- 18.550	86.039	- 15.550	85.949	- 3.550	93.949	470+960.00	94.056	85.327	8.728	9,562,033.871	519,832.612	3.550	93.949									5.049	92.950	11.812	88.441						
- 19.529	85.216	- 5.049	92.780									- 18.550	85.869	- 15.550	85.779	- 3.550	93.779	470+980.00	93.886	85.000	8.886	9,562,045.869	519,848.614	3.550	93.779									5.049	92.780	14.638	86.387						
- 14.153	86.549	- 5.049	92.617													- 3.550	93.617	471+000.00	93.723	86.194	7.530	9,562,057.866	519,864.615	3.550	93.617	15.550	85.617	- 18.550	85.707					5.049	92.617	19.677	84.956						
- 6.533	91.473	- 5.049	92.462													- 3.550	93.462	471+020.00	93.568	91.779	1.789	9,562,069.864	519,880.617	3.550	93.462									5.049	92.462	8.298	90.297						
- 5.751	94.794															- 4.250	93.293	471+040.00	93.420	95.013	- 1.592	9,562,081.862	519,896.619	4.250	93.293												5.607	94.650					
- 6.398	95.301															- 4.250	93.152	471+060.00	93.280	95.652	- 2.372	9,562,093.859	519,912.621	4.250	93.152												6.341	95.244					
- 5.864	94.633															- 4.250	93.019	471+080.00	93.147	94.847	- 1.701	9,562,105.857	519,928.622	4.250	93.019												5.714	94.483					
- 5.364	94.007															- 4.250	92.893	471+100.00	93.020	94.209	- 1.189	9,562,117.854	519,944.624	4.250	92.893												5.182	93.825					
- 7.074	93.817	- 5.049	91.792													- 3.550	92.792	471+120.00	92.898	93.005	- 0.107	9,562,129.852	519,960.626	3.550	92.792									5.049	91.792	5.500	91.491						
- 5.180	91.800	- 5.049	91.670													- 3.550	92.669	471+140.00	92.776	92.079	0.697	9,562,141.850	519,976.628	3.550	92.669									5.049	91.670	5.858	92.478						
- 7.258	95.533															- 4.250	92.526	471+160.00	92.653	95.756	- 3.102	9,56																					

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																														FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS				PÁGINA									
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																														NUMERAÇÃO DA TLSA				2 / 2									
TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)																														FTN-MVP10-NS4D-0643-001-REV02				REV.									
MVP 10 - KM 450+000 A KM 500+789.330																																		00									
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																																											
NOTA DE SERVIÇO - FERROVIA																																											
LADO ESQUERDO																		LADO DIREITO																									
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca		Cota		Cota		Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset	
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Projeto	Terreno	Vermelha	Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)			
- 14.467	77.800	- 5.049	84.078											- 3.550	85.078	472+380.00	85.184	78.000	7.184	9,563,044.970	520,807.613	3.550	85.078																5.049	84.078	14.136	78.021	
- 13.754	78.153	- 5.049	83.956											- 3.550	84.955	472+400.00	85.062	79.230	5.831	9,563,061.326	520,819.123	3.550	84.955														5.049	83.956	12.066	79.278			
- 13.105	78.463	- 5.049	83.833											- 3.550	84.833	472+420.00	84.939	80.092	4.848	9,563,077.682	520,830.634	3.550	84.833													5.049	83.833	9.828	80.648				
- 11.510	79.404	- 5.049	83.711											- 3.550	84.710	472+440.00	84.817	80.926	3.891	9,563,094.038	520,842.144	3.550	84.710													5.049	83.711	8.476	81.426				
- 10.535	79.932	- 5.049	83.589											- 3.550	84.588	472+460.00	84.695	80.588	4.106	9,563,110.394	520,853.654	3.550	84.588												5.049	83.589	8.362	81.380					
- 6.656	82.395	- 5.049	83.466											- 3.550	84.466	472+480.00	84.572	82.299	2.273	9,563,126.749	520,865.165	3.550	84.466												5.049	83.466	8.042	81.471					
- 6.154	84.448	- 5.049	83.344											- 3.550	84.343	472+500.00	84.450	84.582	- 0.132	9,563,143.105	520,876.675	3.550	84.343											5.049	83.344	5.907	84.201						
- 4.848	84.800													- 4.250	84.202	472+520.00	84.330	85.000	- 0.670	9,563,159.461	520,888.186	4.250	84.202															4.848	84.800				
- 5.496	82.811	- 5.049	83.109											- 3.550	84.109	472+540.00	84.215	82.947	1.269	9,563,175.817	520,899.696	3.550	84.109										5.049	83.109	5.073	83.133							
- 5.651	82.599	- 5.049	83.000											- 3.550	84.000	472+560.00	84.106	83.393	0.713	9,563,192.172	520,911.206	3.550	84.000										5.049	83.000	5.920	83.871							
- 5.291	82.735	- 5.049	82.896											- 3.550	83.896	472+580.00	84.002	83.328	0.675	9,563,208.528	520,922.717	3.550	83.896										5.049	82.896	5.648	83.495							
- 6.561	81.790	- 5.049	82.798											- 3.550	83.798	472+600.00	83.904	82.244	1.660	9,563,224.884	520,934.227	3.550	83.798										5.049	82.798	5.974	82.182							
- 8.799	80.205	- 5.049	82.705											- 3.550	83.704	472+620.00	83.811	80.444	3.367	9,563,241.240	520,945.737	3.550	83.704										5.049	82.705	8.607	80.333							
- 12.206	77.846	- 5.049	82.617											- 3.550	83.617	472+640.00	83.723	79.495	4.228	9,563,257.595	520,957.248	3.550	83.617										5.049	82.617	8.663	80.208							
- 9.825	79.351	- 5.049	82.535											- 3.550	83.534	472+660.00	83.641	80.748	2.892	9,563,273.951	520,968.758	3.550	83.534										5.049	82.535	7.102	81.166							
- 7.951	80.523	- 5.049	82.457											- 3.550	83.457	472+680.00	83.563	82.097	1.467	9,563,290.307	520,980.269	3.550	83.457										5.049	82.457	5.092	82.500							
- 4.993	84.107													- 4.250	83.364	472+700.00	83.492	84.577	- 1.085	9,563,306.663	520,991.779	4.250	83.364													5.705	84.819						
- 6.124	85.171													- 4.250	83.298	472+720.00	83.425	85.308	- 1.883	9,563,323.019	521,003.289	4.250	83.298													5.929	84.976						
- 4.777	83.763													- 4.250	83.236	472+740.00	83.364	83.973	- 0.609	9,563,339.374	521,014.800	4.250	83.236													4.701	83.687						
- 6.300	81.366	- 5.049	82.199											- 3.550	83.199	472+760.00	83.305	81.598	1.707	9,563,355.730	521,026.310	3.550	83.199										5.049	82.199	6.327	81.348							
- 11.073	78.125	- 5.049	82.141											- 3.550	83.140	472+780.00	83.247	78.028	5.219	9,563,372.088	521,037.818	3.550	83.140										5.049	82.141	12.311	77.300							
- 15.383	75.193	- 5.049	82.082											- 3.550	83.082	472+800.00	83.188	74.314	8.874	9,563,388.484	521,049.270	3.550	83.082	15.550	75.082	18.550	75.172						5.049	82.082	21.625	73.121							
- 23.520	71.800	- 5.049	82.024									- 18.550	75.113	- 15.550	75.023	- 3.550	83.023	472+820.00	83.130	72.000	11.130	9,563,404.983	521,060.573	3.550	83.023	15.550	75.023	18.550	75.113				5.049	82.024	23.520	71.800							
- 22.488	72.429	- 5.049	81.965									- 18.550	75.055	- 15.550	74.965	- 3.550	82.965	472+840.00	83.071	73.000	10.071	9,563,421.649	521,071.629	3.550	82.965	15.550	74.965	18.550	75.055				5.049	81.965	21.932	72.800							
- 22.410	72.423	- 5.049	81.907									- 18.550	74.996	- 15.550	74.906	- 3.550	82.906	472+860.00	83.013	73.000	10.013	9,563,438.541	521,082.338	3.550	82.906	15.550	74.906	18.550	74.996				5.049	81.907	21.898	72.764							
- 23.257	71.800	- 5.049	81.848									- 18.550	74.938	- 15.550	74.848	- 3.550	82.848	472+880.00	82.954	72.358	10.596	9,563,455.710	521,092.594	3.550	82.848	15.550	74.848	18.550	74.938				5.049	81.848	21.511	72.964							
- 19.634	74.156	- 5.049	81.790									- 18.550	74.879	- 15.550	74.789	- 3.550	82.789	472+900.00	82.896	75.569	7.326	9,563,473.200	521,102.292	3.550	82.789							5.049	81.790	13.769	75.976								
- 12.930	76.477	- 5.049	81.731											- 3.550	82.731	472+920.00	82.837	77.497	5.340	9,563,491.018																							

TRANSPORTADORA NORDESTINA Logística S.A.																												FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																				
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TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)																												FTN-MVP10-NS4D-0643-001-REV02				2 / 2																
MVP 10 - KM 450+000 A KM 500+789.330																												CONTRATO				REV.																
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																																00																
NOTA DE SERVIÇO - FERROVIA																																																
LADO ESQUERDO																		LADO DIREITO																														
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca	Cota Projeto	Cota Terreno	Cota Vermelha	Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset								
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)					Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)							
- 25.279	83.688											- 24.913	83.323	- 18.913	83.503	- 14.250	78.840	474+140.00	79.267	86.124	- 6.857	9,564,652.237	521,349.015	14.250	78.840	21.191	85.781	27.191	85.601													28.390	86.800					
- 22.925	82.969											- 22.925	82.969	- 18.569	83.100	- 14.250	78.781	474+160.00	79.209	86.198	- 6.989	9,564,666.692	521,362.837	14.250	78.781	22.250	86.781	28.250	86.601													35.253	87.425					
- 23.898	82.429											- 23.898	82.429	- 18.129	82.602	- 14.250	78.723	474+180.00	79.150	86.058	- 6.908	9,564,680.973	521,376.839	14.250	78.723	22.250	86.723	28.250	86.543	29.468	87.760	35.468	87.580											36.139	88.252			
- 23.680	81.914											- 23.454	81.688	- 17.454	81.868	- 14.250	78.664	474+200.00	79.092	85.471	- 6.379	9,564,695.162	521,390.934	14.250	78.664	21.757	86.171	27.757	85.991														29.135	87.369				
- 23.015	81.191											- 22.702	80.877	- 16.702	81.057	- 14.250	78.606	474+220.00	79.033	84.180	- 5.147	9,564,709.331	521,405.049	14.250	78.606	19.715	84.071	25.715	83.891														27.098	85.274				
- 22.562	80.680											- 21.554	79.671	- 15.554	79.851	- 14.250	78.547	474+240.00	78.975	82.588	- 3.613	9,564,723.500	521,419.165	14.250	78.547	17.570	81.867	23.570	81.687														25.443	83.560				
- 16.094	80.333															- 14.250	78.489	474+260.00	78.916	81.203	- 2.287	9,564,737.668	521,433.280	14.250	78.489																				17.272	81.510		
- 4.416	78.892															- 4.250	78.726	474+280.00	78.854	79.205	- 0.351	9,564,751.837	521,447.396	4.250	78.726																					4.598	79.075	
- 6.121	76.963	- 5.049	77.678													- 3.550	78.678	474+300.00	78.784	77.138	1.647	9,564,766.006	521,461.511	3.550	78.678												5.049	77.678	6.443	76.749								
- 9.065	74.924	- 5.049	77.601													- 3.550	78.601	474+320.00	78.707	75.000	3.707	9,564,780.175	521,475.626	3.550	78.601													5.049	77.601	8.780	75.114							
- 8.946	74.919	- 5.049	77.517													- 3.550	78.516	474+340.00	78.623	75.836	2.786	9,564,794.344	521,489.742	3.550	78.516														5.049	77.517	7.013	76.208						
- 8.329	75.238	- 5.049	77.425													- 3.550	78.424	474+360.00	78.531	76.066	2.465	9,564,808.512	521,503.857	3.550	78.424														5.049	77.425	6.400	76.524						
- 9.558	74.319	- 5.049	77.325													- 3.550	78.325	474+380.00	78.431	74.951	3.480	9,564,822.681	521,517.973	3.550	78.325															5.049	77.325	8.608	74.953					
- 10.177	73.800	- 5.049	77.219													- 3.550	78.218	474+400.00	78.325	74.777	3.547	9,564,836.850	521,532.088	3.550	78.218														5.049	77.219	8.143	75.156						
- 7.269	75.625	- 5.049	77.104													- 3.550	78.104	474+420.00	78.210	75.899	2.311	9,564,851.019	521,546.204	3.550	78.104														5.049	77.104	6.579	76.084						
- 5.485	77.418	- 5.049	76.983													- 3.550	77.982	474+440.00	78.089	78.024	0.064	9,564,865.188	521,560.319	3.550	77.982													5.049	76.983	6.212	78.145							
- 4.973	78.555															- 4.250	77.832	474+460.00	77.960	79.111	- 1.152	9,564,879.356	521,574.434	29.250	77.082																					33.136	80.968	
- 29.384	77.080															- 29.250	76.946	474+480.00	77.823	79.558	- 1.735	9,564,893.525	521,588.550	29.250	76.946																						34.284	81.979
- 29.427	76.979															- 29.250	76.802	474+500.00	77.679	79.301	- 1.622	9,564,907.694	521,602.665	29.250	76.802																						33.862	81.414
- 29.579	76.983															- 29.250	76.654	474+520.00	77.531	79.289	- 1.757	9,564,921.863	521,616.781	29.250	76.654																						33.396	80.800
- 4.803	77.810															- 4.250	77.256	474+540.00	77.384	78.498	- 1.115	9,564,936.032	521,630.896	29.250	76.506																						32.333	79.589
- 5.173	76.254	- 5.049	76.130													- 3.550	77.130	474+560.00	77.236	76.884	0.352	9,564,950.200	521,645.012	3.550	77.130														5.049	76.130	6.069	77.150						
- 8.049	73.983	- 5.049	75.983													- 3.550	76.982	474+580.00	77.089	74.578	2.511	9,564,964.369	521,659.127	3.550	76.982														5.049	75.983	7.067	74.637						
- 9.592	72.807	- 5.049	75.835													- 3.550	76.834	474+600.00	76.941	73.267	3.674	9,564,978.563	521,673.217	3.550	76.834														5.049	75.835	8.754	73.365						
- 6.428	74.768	- 5.049	75.687													- 3.550	76.687	474+620.00	76.793	74.796	1.998	9,564,992.856	521,687.207	3.550	76.687														5.049	75.687	5.956	75.082						
- 6.488	76.978	- 5.049	75.540													- 3.550	76.539	474+640.00	76.646	76.217	0.429	9,565,007.329	521,701.010	3.550	76.539														5.049	75.540	5.840	76.330						
- 23.126	79.797															- 19.250	75.920	474+660.00	76.498	78.085	- 1.587	9,565,022.057	521,714.540	19.250	75.920																						23.089	79.759
- 23.277	79.800															- 19.250	75.773	474+680.00	76.350	80.000	- 3.650	9,565,037.113	521,727.704	19.250	75.773																						24.101	80.624
- 21.748	78.123															- 19.250	75.625	474+700.00	76.203	79.540	- 3.338	9,565,052.562	521,740.404	19.250	75.625																						24.730	81.106
- 20.713	76.941															- 19.250	75.478	474+720.00	76.055	79.355	- 3.300	9,565,068.439	521,752.565	19.250	75.478																						24.572	80.800
- 20.756	76.835															- 19.250	75.330	474+740.00	75.907	79.491	- 3.583	9,565,084.730	521,764.164	19.250	75.330																						24.767	80.847
- 20.327	76.260															- 19.250	75.182	474+760.00	75.760	78.344	- 2.585	9,565,101.417	521,775.188	19.250	75.182																						23.627	79.555

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																												FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																				
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TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)																												FTN-MVP10-NS4D-0643-001-REV02				2 / 2																
MVP 10 - KM 450+000 A KM 500+789.330																												CONTRATO				REV.																
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NOTA DE SERVIÇO - FERROVIA																																																
LADO ESQUERDO																		LADO DIREITO																														
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca		Cota		Cota		Cota		Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset				
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Projeto	Terreno	Vermelha	Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)								
- 5.694	75.151	- 5.049	74.506													- 3.550	75.506	474+780.00	75.612	75.650	- 0.038	9,565,118.478	521,785.623	3.550	75.506																	5.049	74.506	6.419	75.876			
- 19.823	75.460															- 19.250	74.887	474+800.00	75.464	76.561	- 1.097	9,565,135.892	521,795.456	19.250	74.887																				20.951	76.588		
- 20.911	76.401															- 19.250	74.739	474+820.00	75.317	78.137	- 2.820	9,565,153.640	521,804.675	19.250	74.739																					28.863	78.172	
- 27.403	76.565															- 26.322	75.484	- 20.322	75.664	- 19.250	74.592	474+840.00	75.169	78.977	- 3.807	9,565,171.692	521,813.283	19.250	74.592	22.006	77.348	28.006	77.168													29.897	79.059	
- 27.755	76.769															- 26.721	75.735	- 20.721	75.915	- 19.250	74.444	474+860.00	75.022	78.927	- 3.906	9,565,189.991	521,821.352	19.250	74.444	22.165	77.359	28.165	77.179													30.115	79.129	
- 27.971	76.838															- 26.686	75.552	- 20.686	75.732	- 19.250	74.296	474+880.00	74.874	78.309	- 3.435	9,565,208.477	521,828.984	19.250	74.296	21.932	76.979	27.932	76.799													29.707	78.573	
- 27.916	76.634															- 26.407	75.126	- 20.407	75.306	- 19.250	74.149	474+900.00	74.726	77.702	- 2.976	9,565,227.098	521,836.281	19.250	74.149	21.483	76.381	27.483	76.201													29.197	77.916	
- 27.280	75.851															- 25.958	74.529	- 19.958	74.709	- 19.250	74.001	474+920.00	74.579	77.259	- 2.680	9,565,245.806	521,843.352	19.250	74.001	20.812	75.563	26.812	75.383													28.473	77.044	
- 27.425	75.849															- 25.775	74.199	- 19.775	74.379	- 19.250	73.853	474+940.00	74.431	77.000	- 2.569	9,565,264.559	521,850.304	19.250	73.853	20.213	74.816	26.213	74.636													27.731	76.155	
- 27.568	75.843															- 25.395	73.671	- 19.395	73.851	- 19.250	73.706	474+960.00	74.283	76.823	- 2.540	9,565,283.321	521,857.232	19.250	73.706																	21.270	75.726	
- 27.691	75.820															- 19.250	73.558	474+980.00	74.136	75.957	- 1.822	9,565,302.083	521,864.160	19.250	73.558																					20.688	74.997	
- 27.881	75.861															- 19.250	73.411	475+000.00	73.988	75.619	- 1.630	9,565,320.844	521,871.089	19.250	73.411																					20.055	74.215	
- 27.363	75.195															- 19.250	73.263	475+020.00	73.840	74.722	- 0.882	9,565,339.606	521,878.017	19.250	73.263																					19.591	73.604	
- 20.630	74.495															- 19.250	73.115	475+040.00	73.693	74.181	- 0.488	9,565,358.368	521,884.945	19.250	73.115																					19.314	73.180	
- 4.608	73.775															- 4.250	73.418	475+060.00	73.545	73.799	- 0.254	9,565,377.129	521,891.873	4.250	73.418																						4.268	73.435
- 6.246	73.488	- 5.049	72.291													- 3.550	73.291	475+080.00	73.397	73.546	- 0.149	9,565,395.891	521,898.801	3.550	73.291															5.049	72.291	5.932	73.174					
- 5.930	73.025	- 5.049	72.144													- 3.550	73.143	475+100.00	73.250	73.161	0.088	9,565,414.653	521,905.729	3.550	73.143																				5.049	72.144	5.774	72.868
- 5.853	72.800	- 5.049	71.996													- 3.550	72.996	475+120.00	73.102	73.000	0.102	9,565,433.414	521,912.658	3.550	72.996																				5.049	71.996	5.853	72.800
- 6.001	72.800	- 5.049	71.849													- 3.550	72.848	475+140.00	72.955	72.964	- 0.010	9,565,452.176	521,919.586	3.550	72.848																				5.049	71.849	5.678	72.477
- 6.184	72.836	- 5.049	71.701													- 3.550	72.700	475+160.00	72.807	72.738	0.069	9,565,470.938	521,926.514	3.550	72.700																				5.049	71.701	5.610	72.261
- 4.518	72.800															- 4.250	72.532	475+180.00	72.659	73.000	- 0.341	9,565,489.699	521,933.442	4.250	72.532																						4.518	72.800
- 4.666	72.800															- 4.250	72.384	475+200.00	72.512	73.000	- 0.488	9,565,508.461	521,940.370	4.250	72.384																						4.666	72.800
- 6.207	72.416	- 5.049	71.258													- 3.550	72.257	475+220.00	72.364	72.530	- 0.166	9,565,527.223	521,947.298	3.550	72.257																				5.049	71.258	5.889	72.098
- 5.623	71.684	- 5.049	71.110													- 3.550	72.110	475+240.00	72.216	71.747	0.469	9,565,545.984	521,954.227	3.550	72.110																				5.049	71.110	5.336	71.397
- 5.214	70.853	- 5.049	70.963													- 3.550	71.962	475+260.00	72.069	70.969	1.100	9,565,564.730	521,961.199	3.550	71.962						</																	

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FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																												NUMERAÇÃO DA TLSA										PÁGINA			
TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)																												FTN-MVP10-NS4D-0643-001-REV02										2 / 2			
MVP 10 - KM 450+000 A KM 500+789.330																												CONTRATO										REV.			
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																																						00			
NOTA DE SERVIÇO - FERROVIA																																									
LADO ESQUERDO																		LADO DIREITO																							
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca		Cota Terreno		Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset	
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)		Projeto		Vermelha	Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)		
- 36.249	58.158															- 34.250	56.159	480+060.00	57.187	58.545	- 1.358	9,569,187.555	524,695.098	34.250	56.159															36.746	58.655
- 35.891	57.800															- 34.250	56.159	480+080.00	57.187	58.326	- 1.139	9,569,205.747	524,703.406	34.250	56.159															36.592	58.502
- 35.891	57.800															- 34.250	56.159	480+100.00	57.187	58.099	- 0.912	9,569,223.940	524,711.714	34.250	56.159															36.743	58.652
- 35.891	57.800															- 34.250	56.159	480+120.00	57.187	58.089	- 0.902	9,569,242.133	524,720.021	34.250	56.159															37.173	59.083
- 35.940	57.850															- 34.250	56.159	480+140.00	57.187	59.391	- 2.204	9,569,260.326	524,728.329	34.250	56.159															37.891	59.800
- 36.052	57.961															- 34.250	56.159	480+160.00	57.187	59.904	- 2.717	9,569,278.519	524,736.637	34.250	56.159															37.891	59.800
- 35.996	57.905															- 34.250	56.159	480+180.00	57.187	59.593	- 2.406	9,569,296.712	524,744.944	34.250	56.159															37.301	59.211
- 35.565	57.474															- 34.250	56.159	480+200.00	57.187	58.859	- 1.672	9,569,314.905	524,753.252	34.250	56.159															36.654	58.563
- 35.161	57.071															- 34.250	56.159	480+220.00	57.187	57.985	- 0.798	9,569,333.098	524,761.560	34.250	56.159															36.054	57.963
- 34.821	56.731															- 34.250	56.159	480+240.00	57.187	57.534	- 0.347	9,569,351.291	524,769.868	34.250	56.159															35.602	57.511
- 5.779	56.811	- 5.049	56.081													- 3.550	57.081	480+260.00	57.187	57.075	0.112	9,569,369.484	524,778.175	3.550	57.081									5.049	56.081	5.863	56.894				
- 5.471	55.800	- 5.049	56.081													- 3.550	57.081	480+280.00	57.187	56.000	1.187	9,569,387.676	524,786.483	3.550	57.081									5.049	56.081	5.471	55.800				
- 5.569	56.600	- 5.049	56.081													- 3.550	57.081	480+300.00	57.187	57.000	0.187	9,569,405.869	524,794.791	3.550	57.081									5.049	56.081	5.768	56.800				
- 35.446	57.356															- 34.250	56.159	480+320.00	57.187	57.890	- 0.703	9,569,424.062	524,803.099	34.250	56.159															35.898	57.808
- 36.126	58.035															- 34.250	56.159	480+340.00	57.187	58.529	- 1.342	9,569,442.256	524,811.405	34.250	56.159															36.515	58.425
- 36.627	58.536															- 34.250	56.159	480+360.00	57.187	59.000	- 1.813	9,569,460.473	524,819.658	34.250	56.159															37.113	59.023
- 36.891	58.800															- 34.250	56.159	480+380.00	57.187	59.000	- 1.813	9,569,478.760	524,827.756	34.250	56.159															37.337	59.246
- 36.891	58.800															- 34.250	56.159	480+400.00	57.187	59.000	- 1.813	9,569,497.162	524,835.591	34.250	56.159															37.422	59.332
- 36.788	58.697															- 34.250	56.159	480+420.00	57.187	58.833	- 1.646	9,569,515.717	524,843.053	34.250	56.159															37.216	59.125
- 35.942	57.851															- 34.250	56.159	480+440.00	57.187	58.437	- 1.250	9,569,534.460	524,850.030	34.250	56.159															36.890	58.800
- 35.135	57.044															- 34.250	56.159	480+460.00	57.187	58.929	- 1.742	9,569,553.415	524,856.407	34.250	56.159															36.890	58.800
- 35.432	57.342															- 34.250	56.159	480+480.00	57.187	59.000	- 1.813	9,569,572.596	524,862.067	34.250	56.159															37.055	58.965
- 35.614	57.524															- 34.250	56.159	480+500.00	57.187	59.062	- 1.875	9,569,592.004	524,866.888	34.250	56.159															37.265	59.174
- 35.963	57.872															- 34.250	56.159	480+520.00	57.187	59.041	- 1.854	9,569,611.623	524,870.765	34.250	56.159															37.466	59.376
- 35.891	57.800															- 34.250	56.159	480+540.00	57.187	58.722	- 1.535	9,569,631.409	524,873.668	34.250	56.159															37.510	59.420
- 35.890	57.800															- 34.250	56.159	480+560.00	57.187	58.258	- 1.071	9,569,651.315	524,875.589	34.250	56.159															37.023	58.932
- 4.990	57.800															- 4.250	57.060	480+580.00	57.187	58.000	- 0.813	9,569,671.291	524,876.524	4.250	57.060															4.990	57.800
- 4.581	57.390															- 4.250	57.060	480+600.00	57.187	57.444	- 0.257	9,569,691.289	524,876.471	4.250	57.060															4.255	57.065
- 5.067	56.099	- 5.049	56.081													- 3.550	57.081	480+620.00	57.187	56.000	1.187	9,569,711.259	524,875.429	3.550	57.081									5.049	56.081	5.117	56.149				
- 5.944	55.484	- 5.049	56.																																						

		FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS	
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS		NUMERAÇÃO DA TISA	PÁGINA
TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)		FTN-MVP10-NS4D-0643-001-REV02	2 / 2
MVP 10 - KM 450+000 A KM 500+789.330		CONTRATO	REV.
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS			00

NOTA DE SERVIÇO - FERROVIA																																									
LADO ESQUERDO																		EIXO					LADO DIREITO																		
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca	Cota Projeto	Cota Terreno	Cota Vermelha	Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset	
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)					Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)
- 8.762	49.183	- 5.049	51.658													- 3.550	52.658	481+820.00	52.764	48.959	3.805	9,570,631.443	524,150.686	3.550	52.658												5.049	51.658	10.140	48.264	
- 8.052	49.551	- 5.049	51.553													- 3.550	52.552	481+840.00	52.659	49.196	3.463	9,570,643.995	524,135.115	3.550	52.552											5.049	51.553	9.958	48.281		
- 7.510	49.807	- 5.049	51.447													- 3.550	52.447	481+860.00	52.553	49.529	3.025	9,570,656.546	524,119.544	3.550	52.447											5.049	51.447	9.110	48.740		
- 6.853	50.139	- 5.049	51.342													- 3.550	52.342	481+880.00	52.448	50.000	2.448	9,570,669.097	524,103.973	3.550	52.342											5.049	51.342	8.218	49.230		
- 6.173	50.487	- 5.049	51.237													- 3.550	52.236	481+900.00	52.343	50.403	1.940	9,570,681.649	524,088.401	3.550	52.236											5.049	51.237	7.089	49.877		
- 6.059	50.458	- 5.049	51.131													- 3.550	52.131	481+920.00	52.237	50.399	1.839	9,570,694.200	524,072.830	3.550	52.131											5.049	51.131	6.912	49.890		
- 6.502	50.058	- 5.049	51.026													- 3.550	52.026	481+940.00	52.132	50.008	2.124	9,570,706.751	524,057.259	3.550	52.026											5.049	51.026	7.339	49.500		
- 7.156	49.516	- 5.049	50.921													- 3.550	51.920	481+960.00	52.027	49.405	2.622	9,570,719.303	524,041.688	3.550	51.920											5.049	50.921	8.140	48.861		
- 7.673	49.067	- 5.049	50.816													- 3.550	51.815	481+980.00	51.922	48.956	2.966	9,570,731.854	524,026.116	3.550	51.815											5.049	50.816	8.781	48.327		
- 7.976	48.759	- 5.049	50.710													- 3.550	51.710	482+000.00	51.816	48.580	3.236	9,570,744.405	524,010.545	3.550	51.710											5.049	50.710	9.230	47.923		
- 8.028	48.619	- 5.049	50.605													- 3.550	51.604	482+020.00	51.711	48.512	3.199	9,570,756.957	523,994.974	3.550	51.604											5.049	50.605	9.017	47.960		
- 8.159	48.427	- 5.049	50.500													- 3.550	51.499	482+040.00	51.606	48.362	3.243	9,570,769.508	523,979.403	3.550	51.499											5.049	50.500	9.019	47.853		
- 8.362	48.186	- 5.049	50.394													- 3.550	51.394	482+060.00	51.500	48.104	3.396	9,570,782.059	523,963.831	3.550	51.394											5.049	50.394	9.203	47.625		
- 8.313	48.113	- 5.049	50.289													- 3.550	51.289	482+080.00	51.395	48.024	3.371	9,570,794.611	523,948.260	3.550	51.289											5.049	50.289	9.150	47.555		
- 8.260	48.043	- 5.049	50.184													- 3.550	51.183	482+100.00	51.290	47.969	3.321	9,570,807.162	523,932.689	3.550	51.183											5.049	50.184	9.053	47.515		
- 8.278	47.926	- 5.049	50.078													- 3.550	51.078	482+120.00	51.184	47.895	3.290	9,570,819.714	523,917.118	3.550	51.078											5.049	50.078	9.004	47.442		
- 8.479	47.687	- 5.049	49.973													- 3.550	50.973	482+140.00	51.079	47.693	3.386	9,570,832.275	523,901.555	3.550	50.973											5.049	49.973	9.092	47.278		
- 8.623	47.485	- 5.049	49.868													- 3.550	50.867	482+160.00	50.974	47.507	3.466	9,570,844.916	523,886.056	3.550	50.867											5.049	49.868	9.190	47.107		
- 8.872	47.214	- 5.049	49.762													- 3.550	50.762	482+180.00	50.868	47.280	3.588	9,570,857.726	523,870.697	3.550	50.762											5.049	49.762	9.322	46.914		
- 9.278	46.838	- 5.049	49.657													- 3.550	50.657	482+200.00	50.763	46.895	3.868	9,570,870.790	523,855.554	3.550	50.657											5.049	49.657	9.757	46.519		
- 9.580	46.531	- 5.049	49.552													- 3.550	50.551	482+220.00	50.658	46.639	4.019	9,570,884.156	523,840.677	3.550	50.551											5.049	49.552	9.913	46.310		
- 9.789	46.287	- 5.049	49.447													- 3.550	50.446	482+240.00	50.553	46.349	4.204	9,570,897.823	523,826.075	3.550	50.446											5.049	49.447	10.145	46.049		
- 9.905	46.104	- 5.049	49.341													- 3.550	50.341	482+260.00	50.447	46.155	4.293	9,570,911.784	523,811.754	3.550	50.341											5.049	49.341	10.361	45.800		
- 10.163	45.827	- 5.049	49.236													- 3.550	50.235	482+280.00	50.342	45.960	4.382	9,570,926.033	523,797.721	3.550	50.235											5.049	49.236	10.314	45.726		
- 10.467	45.518	- 5.049	49.131													- 3.550	50.130	482+300.00	50.237	45.592	4.645	9,570,940.565	523,783.980	3.550	50.130											5.049	49.131	10.755	45.327		
- 10.717	45.247	- 5.049	49.025													- 3.550	50.025	482+320.00	50.131	45.297	4.835	9,570,955.375	523,770.539	3.550	50.025											5.049	49.025	11.163	44.949		
- 10.977	44.968	- 5.049	48.920													- 3.550	49.920	482+340.00	50.026	45.048	4.978	9,570,970.454	523,757.402	3.550	49.920											5.049	48.920	11.229	44.800		
- 11.071	44.800	- 5.049	48.815													- 3.550	49.814	482+360.00	49.921	45.000	4.921	9,570,985.798	523,744.574	3.550	49.814											5.049	48.815	11.071	44.800		
- 12.142	43.981	- 5.049	48.709													- 3.550	49.709	482+380.00	49.815	44.351	5.464	9,571,001.400	523,732.061	3.550	49.709											5.049	48.709	11.598	44.344		
- 12.248	43.805	- 5.049	48.604													- 3.550	49.604	482+400.00	49.710	44.000	5.710	9,571,017.238	523,719.849	3.550	49.604											5.049	48.604	12.255	43.800		
- 10.480	44.878	- 5.049	48.499													- 3.550	49.498	482+420.00	49.605	44.785	4.820	9,571,033.245	523,707.858	3.550	49.498											5.049	48.499	11.293	44.337		
- 9.464	45.450	- 5.049	48.393													- 3.550	49.393	482+440.00	49.499	45.440	4.059	9,571,049.350	523,695.999	3.550	49.393											5.049	48.393	10.215	44.950		
- 8.187	46.197	- 5.049	48.288													- 3.550	49.288	482+460.00	49.394	46.375	3.019	9,571,065.484	523,684.180	3.550	49.288											5.049	48.288	8.403	46.052		
- 6.232	47.395	- 5.049	48.183													- 3.550	49.182	482+480.00	49.289	47.548	1.740	9,571,081.619	523,672.362	3.550	49.182											5.049	48.183	6.354	47.313		
- 4.843	49.649															- 4.250	49.056	482+500.00	49.184	49.910	0.727	9,571,097.754																			

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FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																												FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS													
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																												NUMERAÇÃO DA TLSA				PÁGINA									
TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)																												FTN-MVP10-NS4D-0643-001-REV02				2 / 2									
MVP 10 - KM 450+000 A KM 500+789.330																												CONTRATO				REV.									
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																																00									
NOTA DE SERVIÇO - FERROVIA																																									
LADO ESQUERDO														EIXO						LADO DIREITO																					
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca	Cota Projeto	Cota Terreno	Cota Vermelha	Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset	
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)					Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)
- 5.186	39.803	- 5.049	39.666													- 3.550	40.666	486+700.00	40.772	39.966	0.807	9,574,820.064	521,879.425	3.550	40.666											5.049	39.666	5.110	39.727		
- 5.777	39.258	- 5.049	39.743													- 3.550	40.743	486+720.00	40.849	39.524	1.325	9,574,834.209	521,865.286	3.550	40.743											5.049	39.743	5.687	39.318		
- 6.580	38.800	- 5.049	39.820													- 3.550	40.820	486+740.00	40.926	39.040	1.886	9,574,848.079	521,850.878	3.550	40.820											5.049	39.820	6.462	38.878		
- 6.956	38.626	- 5.049	39.897													- 3.550	40.897	486+760.00	41.003	38.646	2.357	9,574,861.755	521,836.284	3.550	40.897											5.049	39.897	7.518	38.251		
- 6.838	38.781	- 5.049	39.974													- 3.550	40.974	486+780.00	41.080	38.836	2.245	9,574,875.320	521,821.587	3.550	40.974											5.049	39.974	7.291	38.480		
- 6.842	38.856	- 5.049	40.051													- 3.550	41.051	486+800.00	41.157	39.100	2.057	9,574,888.853	521,806.861	3.550	41.051											5.049	40.051	6.726	38.933		
- 6.476	39.177	- 5.049	40.128													- 3.550	41.128	486+820.00	41.234	39.401	1.833	9,574,902.386	521,792.135	3.550	41.128											5.049	40.128	6.369	39.249		
- 6.054	39.535	- 5.049	40.205													- 3.550	41.205	486+840.00	41.311	39.739	1.572	9,574,915.919	521,777.409	3.550	41.205											5.049	40.205	6.043	39.543		
- 5.388	40.056	- 5.049	40.282													- 3.550	41.282	486+860.00	41.388	40.199	1.189	9,574,929.452	521,762.683	3.550	41.282											5.049	40.282	5.648	39.883		
- 5.587	40.896	- 5.049	40.359													- 3.550	41.359	486+880.00	41.465	41.187	0.278	9,574,943.002	521,747.973	3.550	41.359											5.049	40.359	5.753	41.063		
- 4.561	41.726															- 4.250	41.415	486+900.00	41.542	41.963	- 0.421	9,574,956.642	521,733.345	4.250	41.415													4.636	41.801		
- 5.293	42.534															- 4.250	41.491	486+920.00	41.619	42.780	- 1.161	9,574,970.455	521,718.882	4.250	41.491													5.371	42.613		
- 5.751	43.069															- 4.250	41.568	486+940.00	41.696	43.309	- 1.613	9,574,984.523	521,704.667	4.250	41.568													5.831	43.149		
- 12.069	43.285											- 10.326	41.541	- 4.326	41.721	- 4.250	41.645	486+960.00	41.773	43.638	- 1.865	9,574,998.922	521,690.788	4.250	41.645	4.437	41.832	10.437	41.652									12.380	43.595		
- 6.100	43.573															- 4.250	41.722	486+980.00	41.850	43.760	- 1.910	9,575,013.723	521,677.338	4.250	41.722													6.020	43.493		
- 5.565	43.114															- 4.250	41.799	487+000.00	41.927	43.291	- 1.364	9,575,028.988	521,664.418	4.250	41.799													5.559	43.109		
- 4.924	42.550															- 4.250	41.876	487+020.00	42.004	42.859	- 0.856	9,575,044.770	521,652.135	4.250	41.876													5.147	42.773		
- 6.233	42.158	- 5.049	40.975													- 3.550	41.974	487+040.00	42.081	42.000	0.081	9,575,061.108	521,640.602	3.550	41.974											5.049	40.975	5.874	41.800		
- 4.595	42.375															- 4.250	42.030	487+060.00	42.158	42.370	- 0.212	9,575,077.995	521,629.890	4.250	42.030													4.331	41.976		
- 6.469	42.549	- 5.049	41.129													- 3.550	42.128	487+080.00	42.235	42.431	- 0.196	9,575,095.390	521,620.025	3.550	42.128											5.049	41.129	5.863	41.942		
- 4.660	42.594															- 4.250	42.184	487+100.00	42.312	42.569	- 0.257	9,575,113.252	521,611.033	4.250	42.184												4.250	42.184			
- 4.624	42.636															- 4.250	42.261	487+120.00	42.389	42.655	- 0.267	9,575,131.537	521,602.934	4.250	42.261												4.277	42.289			
- 6.124	42.435	- 5.049	41.360													- 3.550	42.359	487+140.00	42.466	42.393	0.072	9,575,150.199	521,595.748	3.550	42.359										5.049	41.360	5.650	41.961			
- 5.646	42.034	- 5.049	41.437													- 3.550	42.436	487+160.00	42.543	42.099	0.444	9,575,169.194	521,589.494	3.550	42.436										5.049	41.437	5.322	41.709			
- 5.217	41.682	- 5.049	41.514													- 3.550	42.513	487+180.00	42.620	41.849	0.770	9,575,188.475	521,584.186	3.550	42.513										5.049	41.514	5.073	41.537			
- 5.139	41.531	- 5.049	41.591													- 3.550	42.590	487+200.00	42.697	41.664	1.033	9,575,207.995	521,579.838	3.550	42.590										5.049	41.591	5.378	41.371			
- 5.718	41.222	- 5.049	41.668													- 3.550	42.667	487+220.00	42.774	41.200	1.573	9,575,227.705	521,576.460	3.550	42.667										5.049	41.668	6.351	40.800			
- 5.259	41.605	- 5.049	41.745													- 3.550	42.744	487+240.00	42.851	41.649	1.201	9,575,247.559	521,574.060	3.550	42.744										5.049	41.745	5.705	41.308			
- 5.142	41.760	- 5.049	41.822													- 3.550	42.821	487+260.00	42.928	41.817	1.110	9,575,267.506	521,572.645	3.550	42.821										5.049	41.822	5.588	41.463			
- 5.622	41.516	- 5.049	41.899													- 3.550	42.898	487+280.00	43.005	41.654	1.350	9,575,287.500	521,572.217	3.550	42.898										5.049	41.899	5.810	41.392			
- 8.313	39.800	- 5.049	41.976													- 3.550	42.975	487+300.00	43.082	40.276	2.806	9,575,307.490	521,572.778	3.550	42.975										5.049	41.976	7.144	40.579			
- 8.027	40.067	- 5.049	42.052													- 3.550	43.052	487+320.00	43.158	40.010	3.148	9,575,327.428	521,574.326	3.550	43.052										5.049	42.052	8.428	39.800			
- 6.239	41.337	- 5.049	42.129													- 3.550	43.129	487+340.00	43.235	41.454	1.781	9,575,347.265	521,576.858	3.550	43.129										5.049	42.129	6.497	41.164			
- 5.287	42.048	- 5.049	42.206													- 3.550	43.206	487+360.00	43.312	42.144	1.169	9,575,366.953	521,580.367	3.550	43.206										5.049	42.206	5.595	41.843			
- 5.532	42.767	- 5.049	42.283													- 3.550	43.283	487+380.00	43.389	42.900	0.489	9,575,386.443	521,584.845	3.550	43.283									5.049	42.283	5.389	42.622				

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																																FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS				NUMERAÇÃO DA TLSA		PÁGINA								
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NOTA DE SERVIÇO - FERROVIA																																														
LADO ESQUERDO																		LADO DIREITO																												
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca		Cota		Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset						
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)		Projeto	Terreno	Vermelha	Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)							
- 4.573	43.662													- 4.250	43.339	487+400.00	43.466	43.782	- 0.316	9,575,405.688	521,590.280	4.250	43.339																			4.416	43.505			
- 5.609	44.775													- 4.250	43.416	487+420.00	43.543	44.754	- 1.210	9,575,424.641	521,596.661	4.250	43.416																				5.208	44.373		
- 5.578	44.821													- 4.250	43.493	487+440.00	43.620	44.795	- 1.174	9,575,443.255	521,603.970	4.250	43.493																				5.123	44.366		
- 5.406	44.726													- 4.250	43.570	487+460.00	43.697	44.750	- 1.052	9,575,461.495	521,612.169	4.250	43.570																				5.104	44.424		
- 5.167	44.564													- 4.250	43.647	487+480.00	43.774	44.660	- 0.886	9,575,479.363	521,621.151	4.250	43.647																				4.987	44.384		
- 4.585	44.059													- 4.250	43.724	487+500.00	43.851	44.166	- 0.315	9,575,496.883	521,630.796	4.250	43.724																				4.407	43.881		
- 5.746	43.519	- 5.049	42.822											- 3.550	43.822	487+520.00	43.928	43.522	0.406	9,575,514.089	521,640.991	3.550	43.822														5.049	42.822	5.400	43.173						
- 5.279	43.129	- 5.049	42.899											- 3.550	43.899	487+540.00	44.005	43.042	0.964	9,575,531.025	521,651.627	3.550	43.899																			5.049	42.899			
- 5.318	42.797	- 5.049	42.976											- 3.550	43.976	487+560.00	44.082	42.832	1.250	9,575,547.746	521,662.601	3.550	43.976																			5.049	42.976			
- 6.147	42.321	- 5.049	43.053											- 3.550	44.053	487+580.00	44.159	42.374	1.785	9,575,564.309	521,673.811	3.550	44.053																			5.049	43.053			
- 7.044	41.800	- 5.049	43.130											- 3.550	44.130	487+600.00	44.236	42.000	2.236	9,575,580.777	521,685.159	3.550	44.130																			5.049	43.130			
- 6.729	42.088	- 5.049	43.207											- 3.550	44.207	487+620.00	44.313	42.096	2.218	9,575,597.214	521,696.553	3.550	44.207																			5.049	43.207			
- 5.710	42.843	- 5.049	43.284											- 3.550	44.284	487+640.00	44.390	43.129	1.261	9,575,613.650	521,707.949	3.550	44.284																			5.049	43.284			
- 5.488	43.800	- 5.049	43.361											- 3.550	44.361	487+660.00	44.467	44.159	0.308	9,575,630.086	521,719.344	3.550	44.361																			5.049	43.361			
- 30.636	45.053													- 29.250	43.667	487+680.00	44.544	45.079	- 0.535	9,575,646.522	521,730.739	29.250	43.667																					30.776	45.192	
- 31.763	46.257													- 29.250	43.744	487+700.00	44.621	46.038	- 1.417	9,575,662.959	521,742.134	29.250	43.744																					31.570	46.063	
- 32.679	47.249													- 29.250	43.820	487+720.00	44.698	46.849	- 2.151	9,575,679.395	521,753.530	29.250	43.820																						32.230	46.800
- 33.403	48.050													- 29.250	43.897	487+740.00	44.775	47.572	- 2.797	9,575,695.831	521,764.925	29.250	43.897																						32.160	46.807
- 34.014	48.738													- 29.250	43.974	487+760.00	44.852	48.063	- 3.211	9,575,712.267	521,776.320	29.250	43.974																						31.859	46.584
- 34.326	49.127													- 29.250	44.051	487+780.00	44.929	48.233	- 3.304	9,575,728.705	521,787.713	29.250	44.051																						31.759	46.560
- 34.532	49.410													- 29.250	44.128	487+800.00	45.006	48.740	- 3.734	9,575,745.180	521,799.053	29.250	44.128																						32.072	46.950
- 34.845	49.800													- 29.250	44.205	487+820.00	45.083	49.190	- 4.107	9,575,761.755	521,810.244	29.250	44.205																						32.378	47.333
- 34.924	49.956													- 29.250	44.282	487+840.00	45.160	49.539	- 4.379	9,575,778.494	521,821.190	29.250	44.282																						32.877	47.910
- 34.691	49.800													- 29.250	44.359	487+860.00	45.237	49.631	- 4.394	9,575,795.454	521,831.788	29.250	44.359																						33.336	48.445
- 34.272	49.458													- 29.250	44.436	487+880.00	45.314	49.903	- 4.589	9,575,812.688	521,841.935	29.250	44.436																						33.967	49.153
- 33.516	48.780													- 29.250	44.513	487+900.00	45.391	49.466	- 4.075	9,575,830.238	521,851.524	29.250	44.513																						34.633	49.896
- 32.402	47.742													- 29.250	44.590	487+920.00	45.468	48.808	- 3.340	9,575,848.112	521,860.496	29.250	44.590																						34.918	50.258
- 30.827	46.244													- 29.250	44.667	487+940.00	45.545	48.453	- 2.909	9,575,866.287	521,868.839	29.250	44.667																							

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NOTA DE SERVIÇO - FERROVIA																																										
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Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca	Cota Projeto	Cota Terreno	Cota Vermelha	Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset		
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)					Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)
- 13.911	61.556	- 5.049	67.465													- 3.550	68.464	491+560.00	68.571	61.625	6.946	9,578,816.936	520,160.030		3.550	68.464													5.049	67.465	14.020	61.484
- 14.724	60.885	- 5.049	67.335													- 3.550	68.335	491+580.00	68.441	61.207	7.234	9,578,836.792	520,162.429		3.550	68.335													5.049	67.335	14.072	61.320
- 14.635	60.800	- 5.049	67.190													- 3.550	68.190	491+600.00	68.296	61.000	7.296	9,578,856.647	520,164.829		3.550	68.190													5.049	67.190	14.394	60.960
- 14.771	60.549	- 5.049	67.030													- 3.550	68.029	491+620.00	68.136	61.333	6.803	9,578,876.503	520,167.229		3.550	68.029												5.049	67.030	13.300	61.529	
- 13.881	60.966	- 5.049	66.854													- 3.550	67.853	491+640.00	67.960	61.712	6.247	9,578,896.358	520,169.629		3.550	67.853												5.049	66.854	12.424	61.937	
- 13.206	61.224	- 5.049	66.662													- 3.550	67.662	491+660.00	67.768	61.826	5.942	9,578,916.214	520,172.029		3.550	67.662												5.049	66.662	12.102	61.960	
- 12.707	61.359	- 5.049	66.464													- 3.550	67.464	491+680.00	67.570	61.831	5.739	9,578,936.069	520,174.429		3.550	67.464												5.049	66.464	11.895	61.900	
- 12.576	61.248	- 5.049	66.266													- 3.550	67.265	491+700.00	67.372	61.715	5.657	9,578,955.925	520,176.829		3.550	67.265												5.049	66.266	11.819	61.753	
- 12.460	61.127	- 5.049	66.068													- 3.550	67.067	491+720.00	67.174	61.585	5.589	9,578,975.780	520,179.229		3.550	67.067												5.049	66.068	11.703	61.632	
- 11.932	61.281	- 5.049	65.870													- 3.550	66.869	491+740.00	66.976	61.788	5.187	9,578,995.636	520,181.629		3.550	66.869												5.049	65.870	11.154	61.800	
- 11.517	61.360	- 5.049	65.671													- 3.550	66.671	491+760.00	66.777	61.920	4.858	9,579,015.491	520,184.028		3.550	66.671											5.049	65.671	10.580	61.984		
- 11.989	60.847	- 5.049	65.473													- 3.550	66.473	491+780.00	66.579	61.383	5.197	9,579,035.347	520,186.428		3.550	66.473												5.049	65.473	11.554	61.137	
- 11.406	61.037	- 5.049	65.275													- 3.550	66.275	491+800.00	66.381	61.464	4.917	9,579,055.202	520,188.828		3.550	66.275											5.049	65.275	10.577	61.590		
- 9.713	61.968	- 5.049	65.077													- 3.550	66.077	491+820.00	66.183	62.446	3.737	9,579,075.058	520,191.228		3.550	66.077											5.049	65.077	8.992	62.449		
- 8.851	62.344	- 5.049	64.879													- 3.550	65.878	491+840.00	65.985	62.835	3.150	9,579,094.913	520,193.628		3.550	65.878										5.049	64.879	8.002	62.910			
- 8.370	62.467	- 5.049	64.681													- 3.550	65.680	491+860.00	65.787	62.946	2.841	9,579,114.769	520,196.028		3.550	65.680										5.049	64.681	7.536	63.023			
- 8.125	62.432	- 5.049	64.483													- 3.550	65.482	491+880.00	65.589	62.865	2.724	9,579,134.624	520,198.428		3.550	65.482										5.049	64.483	7.413	62.907			
- 8.158	62.212	- 5.049	64.284													- 3.550	65.284	491+900.00	65.390	62.663	2.728	9,579,154.480	520,200.828		3.550	65.284										5.049	64.284	7.389	62.725			
- 8.271	61.938	- 5.049	64.086													- 3.550	65.086	491+920.00	65.192	62.483	2.709	9,579,174.335	520,203.228		3.550	65.086										5.049	64.086	7.344	62.557			
- 8.435	61.631	- 5.049	63.888													- 3.550	64.888	491+940.00	64.994	62.109	2.886	9,579,194.191	520,205.628		3.550	64.888										5.049	63.888	7.494	62.258			
- 8.980	61.070	- 5.049	63.690													- 3.550	64.690	491+960.00	64.796	61.767	3.029	9,579,214.046	520,208.027		3.550	64.690										5.049	63.690	7.416	62.112			
- 7.850	61.625	- 5.049	63.492													- 3.550	64.491	491+980.00	64.598	62.205	2.393	9,579,233.902	520,210.427		3.550	64.491									5.049	63.492	6.591	62.464				
- 7.376	61.742	- 5.049	63.294													- 3.550	64.293	492+000.00	64.400	62.302	2.097	9,579,253.757	520,212.827		3.550	64.293									5.049	63.294	6.372	62.412				
- 7.319	61.583	- 5.049	63.096													- 3.550	64.095	492+020.00	64.202	62.151	2.050	9,579,273.614	520,215.218		3.550	64.095									5.049	63.096	6.421	62.181				
- 8.195	60.800	- 5.049	62.897													- 3.550	63.897	492+040.00	64.003	61.118	2.886	9,579,293.481	520,217.517		3.550	63.897									5.049	62.897	6.695	61.800				
- 6.159	61.959	- 5.049	62.699													- 3.550	63.699	492+060.00	63.805	62.332	1.473	9,579,313.371	520,219.611		3.550	63.699								5.049	62.699	5.724	62.250					
- 5.336	62.310	- 5.049	62.501													- 3.550	63.501	492+080.00	63.607	62.702	0.905	9,579,333.292	520,221.381		3.550	63.501								5.049	62.501	5.239	62.691					
- 5.267	62.521	- 5.049	62.303													- 3.550	63.303	492+100.00	63.409	62.939	0.470	9,579,353.247	520,222.713		3.550	63.303								5.049	62.303	5.738	62.992					
- 5.562	62.617	- 5.049	62.105													- 3.550	63.104	492+120.00	63.211	63.047	0.164	9,579,373.232	520,223.488		3.550	63.104								5.049	62.105	6.091	63.146					
- 4.286	62.861															- 4.250	62.885	492+140.00	63.013	63.252	0.240	9,579,393.230	520,223.591		4.250	62.885												4.635	63.270			
- 4.712	63.149															- 4.250	62.687	492+160.00	62.815	63.520	0.705	9,579,413.217	520,222.905		4.250	62.687													5.065	63.503		
- 4.966	63.205															- 4.250	62.489	492+180.00	62.616	63.557	0.940	9,579,433.152	520,221.315		4.250	62.489													5.275	63.514		
- 4.981	63.022															- 4.250	62.291	492+200.00	62.418	63.363	0.945	9,579,452.984	520,218.748		4.250	62.291													5.273	63.314		
- 4.979	62.822															- 4.250	62.093	492+220.00	62.220	63.078	0.857	9,579,472.666	520,215.205		4.250	62.093													5.096	62.939		
- 4.515	62.160															- 4.250	61.895	492+240.00	62.022	62.437	0.415	9,579,492.148	520,210.693		4.250	61.895													4.690	62.335		
- 5.991	61.660	- 5.049	60.718													- 3.550	61.717	492+260.00	61.824	61.737	0.087	9,579,511.384	520,205.223		3.550	61.717									5.049	60.718	5.786	61.454				
- 5.329	60.800	- 5.049	60.520													- 3.550	61.519	492+280.00	61.626	61.000	0.626	9,579,530.325	520,198.810		3.550	61.519								5.049	60.520	5.427	60.898					
- 5.778	61.051	- 5.049	60.322													- 3.550	61.321	492+300.00	61.428	61.378	0.050	9,579,548.927	520,191.468		3.550	61.321								5.049	60.322	6.072	61.345					
- 4.917	61.769															- 4.250	61.102	492+320.00	61.229	62.000	0.771	9,579,567.143	520,183.216		4.250	61.102												4.948	61.800			
- 5.831	62.																																									

FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																												FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS				PÁGINA							
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																												NUMERAÇÃO DA TLSA				2 / 2							
TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)																												FTN-MVP10-NS4D-0643-001-REV02				REV.							
MVP 10 - KM 450+000 A KM 500+789.330																																00							
FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																																							
NOTA DE SERVIÇO - FERROVIA																																							
LADO ESQUERDO																		LADO DIREITO																					
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		EIXO		Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset	
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Estaca	Cota Projeto	Cota Terreno	Cota Vermelha	Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)		
- 6.123	51.147	- 5.049	51.863													- 3.550	52.862	493+300.00	52.969	51.454	1.515	9,580,232.145	519,468.830	3.550	52.862											5.049	51.863	6.620	50.816
- 5.915	51.250	- 5.049	51.828													- 3.550	52.827	493+320.00	52.934	51.591	1.343	9,580,245.220	519,453.696	3.550	52.827											5.049	51.828	5.520	51.514
- 5.928	51.206	- 5.049	51.793													- 3.550	52.792	493+340.00	52.899	51.611	1.288	9,580,258.328	519,438.590	3.550	52.792											5.049	51.793	5.346	51.595
- 5.127	51.706	- 5.049	51.757													- 3.550	52.757	493+360.00	52.863	52.000	0.863	9,580,271.552	519,423.586	3.550	52.757											5.049	51.757	5.103	51.811
- 5.545	52.218	- 5.049	51.722													- 3.550	52.722	493+380.00	52.828	52.483	0.346	9,580,284.977	519,408.762	3.550	52.722											5.049	51.722	5.637	52.310
- 5.995	52.633	- 5.049	51.687													- 3.550	52.687	493+400.00	52.793	52.879	- 0.085	9,580,298.687	519,394.201	3.550	52.687											5.049	51.687	6.115	52.753
- 4.419	52.800															- 4.250	52.631	493+420.00	52.758	53.112	- 0.354	9,580,312.758	519,379.989	4.250	52.631													4.675	53.056
- 4.488	52.834															- 4.250	52.596	493+440.00	52.723	53.120	- 0.396	9,580,327.261	519,366.219	4.250	52.596													4.603	52.948
- 5.974	52.507	- 5.049	51.582													- 3.550	52.582	493+460.00	52.688	52.750	- 0.062	9,580,342.261	519,352.992	3.550	52.582											5.049	51.582	6.038	52.570
- 5.581	52.079	- 5.049	51.547													- 3.550	52.546	493+480.00	52.653	52.284	0.369	9,580,357.783	519,340.382	3.550	52.546											5.049	51.547	5.590	52.088
- 5.177	51.640	- 5.049	51.512													- 3.550	52.511	493+500.00	52.618	51.786	0.832	9,580,373.806	519,328.414	3.550	52.511											5.049	51.512	5.110	51.573
- 5.509	51.170	- 5.049	51.477													- 3.550	52.476	493+520.00	52.583	51.316	1.267	9,580,390.302	519,317.109	3.550	52.476											5.049	51.477	5.673	51.061
- 6.100	50.741	- 5.049	51.442													- 3.550	52.441	493+540.00	52.548	50.912	1.636	9,580,407.245	519,306.484	3.550	52.441											5.049	51.442	6.190	50.681
- 6.372	50.525	- 5.049	51.407													- 3.550	52.406	493+560.00	52.513	50.699	1.814	9,580,424.606	519,296.558	3.550	52.406											5.049	51.407	6.479	50.453
- 6.682	50.283	- 5.049	51.371													- 3.550	52.371	493+580.00	52.477	50.479	1.999	9,580,442.357	519,287.347	3.550	52.371											5.049	51.371	6.720	50.258
- 6.994	50.040	- 5.049	51.336													- 3.550	52.336	493+600.00	52.442	50.242	2.201	9,580,460.469	519,278.866	3.550	52.336											5.049	51.336	7.000	50.036
- 7.301	49.800	- 5.049	51.301													- 3.550	52.301	493+620.00	52.407	50.003	2.405	9,580,478.910	519,271.130	3.550	52.301											5.049	51.301	7.295	49.804
- 7.249	49.800	- 5.049	51.266													- 3.550	52.266	493+640.00	52.372	50.000	2.372	9,580,497.652	519,264.151	3.550	52.266											5.049	51.266	7.249	49.800
- 7.196	49.800	- 5.049	51.231													- 3.550	52.231	493+660.00	52.337	50.000	2.337	9,580,516.662	519,257.940	3.550	52.231											5.049	51.231	7.196	49.800
- 7.143	49.800	- 5.049	51.196													- 3.550	52.196	493+680.00	52.302	50.000	2.302	9,580,535.908	519,252.509	3.550	52.196											5.049	51.196	7.143	49.800
- 7.091	49.800	- 5.049	51.161													- 3.550	52.160	493+700.00	52.267	50.000	2.267	9,580,555.361	519,247.866	3.550	52.160											5.049	51.161	7.091	49.800
- 7.038	49.800	- 5.049	51.126													- 3.550	52.125	493+720.00	52.232	50.000	2.232	9,580,574.986	519,244.018	3.550	52.125											5.049	51.126	7.038	49.800
- 6.985	49.800	- 5.049	51.091													- 3.550	52.090	493+740.00	52.197	50.000	2.197	9,580,594.751	519,240.972	3.550	52.090											5.049	51.091	6.913	49.848
- 6.933	49.800	- 5.049	51.056													- 3.550	52.055	493+760.00	52.162	50.045	2.117	9,580,614.624	519,238.734	3.550	52.055											5.049	51.056	6.762	49.914
- 6.862	49.812	- 5.049	51.021													- 3.550	52.020	493+780.00	52.127	50.081	2.045	9,580,634.572	519,237.307	3.550	52.020											5.049	51.021	6.657	49.949
- 6.739	49.859	- 5.049	50.986													- 3.550	51.985	493+800.00	52.092	50.126	1.965	9,580,654.561	519,236.693	3.550	51.985											5.049	50.986	6.540	49.992
- 6.759	49.811	- 5.049	50.950																																				

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FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS																										NUMERAÇÃO DA TLSA		PÁGINA	
TRECHO MISSÃO VELHA (CE) - PORTO DO PECÉM (CE)																										FTN-MVP10-NS4D-0643-001-REV02		2 / 2	
MVP 10 - KM 450+000 A KM 500+789.330																										CONTRATO		REV.	
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NOTA DE SERVIÇO - FERROVIA																																													
LADO ESQUERDO																EIXO				LADO DIREITO																									
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca	Cota Projeto	Cota Terreno	Cota Vermelha	Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset					
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)					Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)				
- 4.529	48.462															- 4.250	48.183	496+800.00	48.310	48.814	- 0.504	9,583,308.781	518,238.918	4.250	48.183																4.861	48.793			
- 6.544	50.456															- 4.250	48.162	496+820.00	48.289	51.055	- 2.766	9,583,324.671	518,226.773	4.250	48.162																	7.320	51.231		
- 14.370	52.081											- 12.599	50.310	- 6.599	50.490	- 4.250	48.141	496+840.00	48.269	53.166	- 4.897	9,583,340.306	518,214.302	4.250	48.141	7.421	51.312	13.421	51.132												16.089	53.800			
- 15.558	53.248											- 14.675	52.365	- 8.675	52.545	- 4.250	48.121	496+860.00	48.248	54.000	- 5.752	9,583,355.619	518,201.437	4.250	48.121	8.800	52.671	14.800	52.491												16.109	53.800			
- 16.142	53.812											- 15.323	52.993	- 9.323	53.173	- 4.250	48.100	496+880.00	48.227	54.000	- 5.773	9,583,370.589	518,188.176	4.250	48.100	9.276	53.126	15.276	52.946													16.130	53.800		
- 16.202	53.851											- 15.392	53.041	- 9.392	53.221	- 4.250	48.079	496+900.00	48.207	54.182	- 5.975	9,583,385.208	518,174.528	4.250	48.079	9.421	53.250	15.421	53.070													16.211	53.860		
- 15.728	53.357											- 14.957	52.586	- 8.957	52.766	- 4.250	48.059	496+920.00	48.186	53.797	- 5.611	9,583,399.464	518,160.502	4.250	48.059	9.209	53.018	15.209	52.838													16.038	53.667		
- 15.204	52.812											- 14.323	51.931	- 8.323	52.111	- 4.250	48.038	496+940.00	48.165	53.206	- 5.040	9,583,413.351	518,146.109	4.250	48.038	8.460	52.248	14.460	52.068													15.442	53.050		
- 14.558	52.145											- 13.554	51.142	- 7.554	51.322	- 4.250	48.017	496+960.00	48.145	52.615	- 4.470	9,583,426.901	518,131.400	4.250	48.017	7.891	51.658	13.891	51.478													14.992	52.579		
- 14.138	51.705											- 13.234	50.801	- 7.234	50.981	- 4.250	47.997	496+980.00	48.124	52.000	- 3.876	9,583,440.195	518,116.457	4.250	47.997	7.317	51.064	13.317	50.884													14.410	51.977		
- 14.043	51.589											- 12.962	50.508	- 6.962	50.688	- 4.250	47.976	497+000.00	48.103	52.000	- 3.897	9,583,453.314	518,101.362	4.250	47.976	7.036	50.762	13.036	50.582													14.257	51.802		
- 13.676	51.201											- 12.623	50.148	- 6.623	50.328	- 4.250	47.955	497+020.00	48.083	51.398	- 3.315	9,583,466.347	518,086.192	4.250	47.955	6.600	50.306	12.600	50.126													13.679	51.204		
- 13.131	50.636											- 12.155	49.660	- 6.155	49.840	- 4.250	47.935	497+040.00	48.062	50.764	- 2.702	9,583,479.366	518,071.009	4.250	47.935	6.162	49.846	12.162	49.666													13.142	50.646		
- 12.602	50.086											- 11.774	49.258	- 5.774	49.438	- 4.250	47.914	497+060.00	48.041	50.252	- 2.211	9,583,492.385	518,055.827	4.250	47.914	5.796	49.640	11.796	49.280													12.605	50.089		
- 12.178	49.641											- 11.350	48.813	- 5.350	48.993	- 4.250	47.893	497+080.00	48.021	49.856	- 1.835	9,583,505.405	518,040.644	4.250	47.893	5.386	49.029	11.386	48.849													12.220	49.683		
- 11.852	49.294											- 10.956	48.399	- 4.956	48.579	- 4.250	47.873	497+100.00	48.000	49.535	- 1.535	9,583,518.424	518,025.462	4.250	47.873	4.961	48.583	10.961	48.403													11.874	49.317		
- 11.565	48.986											- 10.671	48.093	- 4.671	48.273	- 4.250	47.852	497+120.00	47.979	49.168	- 1.189	9,583,531.443	518,010.280	4.250	47.852	4.677	48.279	10.677	48.099													11.579	49.001		
- 5.219	48.800															- 4.250	47.831	497+140.00	47.959	49.000	- 1.041	9,583,544.462	517,995.097	4.250	47.831																		5.219	48.800	
- 5.382	48.943															- 4.250	47.811	497+160.00	47.938	49.076	- 1.138	9,583,557.481	517,979.915	4.250	47.811																		5.250	48.810	
- 4.995	48.534															- 4.250	47.790	497+180.00	47.917	48.615	- 0.697	9,583,570.500	517,964.733	4.250	47.790																			4.760	48.299
- 6.151	47.893	- 5.049	46.791													- 3.550	47.790	497+200.00	47.897	48.006	- 0.109	9,583,583.528	517,949.558	3.550	47.790									5.049	46.791	5.843	47.585								
- 5.186	46.679	- 5.049	46.770													- 3.550	47.770	497+220.00	47.876	46.651	- 1.225	9,583,596.630	517,934.447	3.550	47.770									5.049	46.770	5.790	46.276								
- 6.618	45.704	- 5.049	46.749													- 3.550	47.749	497+240.00	47.855	45.588	- 2.267	9,583,609.894	517,919.479	3.550	47.749									5.049	46.749	7.615	45.039								
- 8.098	44.696	- 5.049	46.729													- 3.550	47.728	497+260.00	47.835	44.600	- 3.234	9,583,623.404	517,904.732	3.550	47.728									5.049	46.729	9.247	43.930								
- 9.350	43.841	- 5.049	46.708													- 3.550	47.708	497+280.00	47.814	44.000	- 3.814	9,583,637.240	517,890.291	3.550	47.708									5.049	46.708	9.411	43.800								
- 9.889	43.461	- 5.049	46.687													- 3.550	47.687	497+300.00	47.793	44.110	- 3.683	9,583,651.445	517,876.212	3.550	47.687									5.049	46.687	8.544	44.358								
- 9.398	43.768	- 5.049	46.667													- 3.550	47.666	497+320.00	47.773	44.362	- 3.410	9,583,666.013	517,862.511	3.550	47.666									5.049	46.667	8.082	44.645								
- 8.632	44.257	- 5.049	46.646													- 3.550	47.646	497+340.00	47.752	44.813	- 2.939	9,583,680.936	517,849.196	3.550	47.646									5.049	46.646	7.616	44.935								
- 8.297	44.460	- 5.049	46.625																																										

TRAN NORDESTINA Logística S.A.																									FERROVIA TRANSNORDESTINA - REVISÃO DOS PROJETOS EXECUTIVOS				
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NOTA DE SERVIÇO - FERROVIA																																											
LADO ESQUERDO																		EIXO					LADO DIREITO																				
Offset		CF		3 Banqueta_Externa		3 Banqueta_Interna		2 Banqueta_Externa		2 Banqueta_Interna		1 Banqueta_Externa		1 Banqueta_Interna		Bordo da Plataforma		Estaca	Cota Projeto	Cota Terreno	Cota Vermelha	Coordenadas		Bordo da Plataforma		1 Banqueta_Interna		1 Banqueta_Externa		2 Banqueta_Interna		2 Banqueta_Externa		3 Banqueta_Interna		3 Banqueta_Externa		CF		Offset			
Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)					Norte (m)	Leste (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)	Afast. (m)	Cota (m)		
- 5.927	44.800	- 5.274	45.235													- 3.550	46.385	498+560.00	46.491	45.000	1.491	9,584,803.163	517,460.536	8.550	46.235															10.049	45.235	10.494	44.939
- 5.291	45.232	- 5.274	45.215													- 3.550	46.364	498+580.00	46.471	45.603	0.868	9,584,822.991	517,457.925	8.550	46.214														10.049	45.215	10.588	45.754	
- 5.348	45.268	- 5.274	45.194													- 3.550	46.344	498+600.00	46.450	45.603	0.847	9,584,842.764	517,454.921	8.550	46.194														10.049	45.194	10.506	45.651	
- 5.336	45.132	- 5.274	45.173													- 3.550	46.323	498+620.00	46.429	45.440	0.989	9,584,862.472	517,451.515	8.550	46.173													10.049	45.173	10.322	45.446		
- 5.518	44.990	- 5.274	45.153													- 3.550	46.302	498+640.00	46.409	45.303	1.105	9,584,882.106	517,447.709	8.550	46.152													10.049	45.153	10.202	45.306		
- 5.764	44.806	- 5.274	45.132													- 3.550	46.282	498+660.00	46.388	45.087	1.301	9,584,901.658	517,443.504	8.550	46.132													10.049	45.132	10.237	45.007		
- 6.184	44.505	- 5.274	45.111													- 3.550	46.261	498+680.00	46.367	44.780	1.588	9,584,921.121	517,438.901	8.550	46.111													10.049	45.111	10.646	44.713		
- 6.733	44.118	- 5.274	45.091													- 3.550	46.240	498+700.00	46.347	44.395	1.952	9,584,940.486	517,433.904	8.550	46.090													10.049	45.091	11.150	44.357		
- 7.104	43.850	- 5.274	45.070													- 3.550	46.220	498+720.00	46.326	44.000	2.326	9,584,959.746	517,428.513	8.550	46.070													10.049	45.070	11.857	43.865		
- 6.336	44.341	- 5.274	45.049													- 3.550	46.199	498+740.00	46.305	44.545	1.760	9,584,978.891	517,422.731	8.550	46.049													10.049	45.049	10.525	44.732		
- 5.869	44.632	- 5.274	45.029													- 3.550	46.178	498+760.00	46.285	44.932	1.352	9,584,997.915	517,416.560	8.550	46.028													10.049	45.029	10.392	44.800		
- 5.655	44.754	- 5.274	45.008													- 3.550	46.158	498+780.00	46.264	45.060	1.204	9,585,016.810	517,410.003	8.550	46.008													10.049	45.008	10.127	45.086		
- 5.280	44.993	- 5.274	44.987													- 3.550	46.137	498+800.00	46.243	45.288	0.956	9,585,035.567	517,403.063	8.550	45.987													10.049	44.987	10.335	45.273		
- 5.502	45.194	- 5.274	44.967													- 3.550	46.116	498+820.00	46.223	45.494	0.728	9,585,054.178	517,395.743	8.550	45.966													10.049	44.967	10.508	45.426		
- 5.564	45.235	- 5.274	44.946													- 3.550	46.096	498+840.00	46.202	45.505	0.697	9,585,072.637	517,388.045	8.550	45.946													10.049	44.946	10.526	45.423		
- 5.527	45.178	- 5.274	44.925													- 3.550	46.075	498+860.00	46.181	45.451	0.731	9,585,090.935	517,379.973	8.550	45.925													10.049	44.925	10.513	45.389		
- 5.504	45.134	- 5.274	44.905													- 3.550	46.054	498+880.00	46.161	45.413	0.748	9,585,109.066	517,371.530	8.550	45.904													10.049	44.905	10.508	45.363		
- 5.575	45.185	- 5.274	44.884													- 3.550	46.034	498+900.00	46.140	45.480	0.660	9,585,127.020	517,362.719	8.550	45.884													10.049	44.884	10.600	45.435		
- 5.623	45.212	- 5.274	44.863													- 3.550	46.013	498+920.00	46.119	45.520	0.599	9,585,144.791	517,353.545	8.550	45.863													10.049	44.863	10.712	45.526		
- 5.704	45.272	- 5.274	44.843													- 3.550	45.992	498+940.00	46.099	45.578	0.520	9,585,162.372	517,344.011	8.550	45.842													10.049	44.843	10.786	45.579		
- 5.767	45.314	- 5.274	44.822													- 3.550	45.972	498+960.00	46.078	45.610	0.468	9,585,179.755	517,334.121	8.550	45.822													10.049	44.822	10.818	45.590		
- 5.748	45.275	- 5.274	44.801													- 3.550	45.951	498+980.00	46.057	45.560	0.497	9,585,196.934	517,323.879	8.550	45.801													10.049	44.801	10.761	45.513		
- 5.609	45.116	- 5.274	44.781													- 3.550	45.930	499+000.00	46.037	45.377	0.659	9,585,213.900	517,313.290	8.550	45.780													10.049	44.781	10.568	45.300		
- 5.484	44.970	- 5.274	44.760													- 3.550	45.910	499+020.00	46.016	45.249	0.767	9,585,230.647	517,302.357	8.550	45.760													10.049	44.760	10.489	45.200		
- 5.492	44.957	- 5.274	44.739													- 3.550	45.889	499+040.00	45.995	45.237	0.759	9,585,247.167	517,291.085	8.550	45.739													10.049	44.739	10.499	45.189		
- 5.484	44.928	- 5.274	44.719													- 3.550	45.868	499+060.00	45.975	45.208	0.767	9,585,263.455	517,279.480	8.550	45.718													10.049	44.719	10.490	45.160		
- 5.443	44.866	- 5.274	44.698													- 3.550	45.848	499+080.00	45.954	45.142	0.812	9,585,279.504	517,267.545	8.550	45.698													10.049	44.698	10.436	45.085		
- 5.319	44.722	- 5.274	44.677													- 3.550	45.827	499+100.00	45.933	45.049	0.884	9,585,295.306	517,255.286	8.550	45.677													10.049	44.677	10.310	44.938		
- 5.300	44.640	- 5.274	44.657													- 3.550	45.806	499+120.00	45.913	45.000	0.913	9,585,310.855	517,242.708	8.550	45.656													10.049	44.657	10.193	44.800		
- 5.816	44.275	- 5.274	44.636													- 3.550	45.786	499+140.00	45.892	44.577	1.315	9,585,326.145	517,229.815	8.550	45.636													10.049	44.636	10.603	44.267		
- 6.865	43.555	- 5.274	44.615													- 3.550	45.765	499+160.00	45.871	43.731	2.140	9,585,341.169	517,216.615	8.550	45.615													10.049	44.615	11.969	43.335		
- 7.580	43.058	- 5.274	44.595													- 3.550	45.744	499+180.00	45.851	43.159	2.692	9,585,355.921	517,203.111	8.550	45.594													10.049	44.595	12.741	42.800		
- 6.258	43.918	- 5.274	44.574													- 3.550	45.724	499+200+																									

